Supplementary Information for

Increasing electricity exports through delivery-informed allocation in distribution networks

Junjie Huang, Tao Yu, Zhenning Pan, Yufeng Wu,
Wencong Xiao, Rankai Zhu, Jie Huang, and Yusheng Xue

An unused award identifies permission not exercised, but not the electricity another allocation could have delivered. This Supplementary Information (SI) develops that distinction and its use in later request ceilings. Part I derives private and total-delivery effects, the completed-interval comparison, and the ceiling update. Part II specifies the controlled allocation rules and physical checks. Part III uses public household histories and network models to examine where requests are reduced, when ceilings are updated, and how strongly they respond. The objective throughout is to deliver more available electricity within the existing network limits.

Part I. Request effects and feedback mechanism

S1 Quantities and Experimental Design

Three comparisons answer different questions. A changed request is compared with a forecast-based request to measure its effect on delivery. A completed allocation is compared with a replay to determine whether unused permission could have supported more delivery. Finally, trajectories with and without feedback test whether that information improves later export. The replay estimates a missed opportunity in the completed interval; the trajectory comparison measures whether acting on it helps when output changes. Tables 1–4 give the quantities and designs for these comparisons. Symbols used only in a derivation are defined alongside it.

Supplementary Table 1: Quantities used in allocation and post-event replay. Power quantities are in MW; t is the allocation interval, written as day and half-hour (d, τ) in the Australian study.

Quantity	Meaning and time of use
$r_i(t)$	Submitted request, made before available export is known.
$r_i^{\mathrm{cap}}(t)$, $r_i^{\mathrm{adm}}(t)$	The same request after the connection bound and then the current feedback ceiling are applied. Allocation uses the admitted request.
$x_i(t)$, $y_i(t)$	Awarded access and completed delivery, respectively. Their difference is the unused award.
$a_i(t)$	Available export before the award limits delivery; realized during the interval.
$a_i^{\mathrm{prof}}(t)$	Australian available-export proxy constructed from public PV and load profiles. It generates modeled delivery.
$a_i^{\mathrm{tel}}(t)$	Post-event record supplied to the replay. It equals $a_i(t)$ in the controlled study and $a_i^{\mathrm{prof}}(t)$ in the primary Australian simulation.
C_i, $\psi_i(t)$	Fixed connection bound and current request ceiling. ECF updates only the latter.
Superscript rpl	An allocation or delivery calculated for the completed-interval comparison.

Supplementary Table 2: Export-access allocation rules used in the controlled study

Rule	Allocation principle	How requests interact	Experimental role
WP	Common proportional scale [1]	Shared scale and binding proxy constraints	One-parameter scaling benchmark
ER	Common absolute reduction [2,3]	Shared reduction level	One-parameter reduction benchmark
EA	Common access cap [1]	Shared cap and capped set	One-parameter cap benchmark
ME	Maximum total authorized access [4,5]	Request bounds and active proxy constraints	Total-access conic rule
FA	Authorized access with a normalized equality term [2,6,7]	Request bounds, proxy constraints, and equality term	Access–equality comparison

Supplementary Table 3: Controlled medium-voltage multi-interval design

Factor	Tested levels
Allocation rule	WP, ER, EA, ME, FA
Controlled feeder	141-bus radial feeder; demand scale 0.70; 30 fixed participant buses
Reactive-power setting q_{pf}	Q0, Q95
AC limits	$0.95 \le v_b \le 1.05$ pu; $\max_{e\in\{\mathrm{fr,to}\}} \lvert S_{\ell,e}\rvert/\bar{S}_{\ell}^{\mathrm{syn}} \le 1$ at both branch ends
Request-choice grid Γ	$\{1.0, 1.1, \ldots, 2.0\}$; smallest maximizer
Static uncertainty σ	0, 0.05, 0.10, 0.20, 0.30
Static normalized shortfall cost ϑ	0, 0.05, 0.10, 0.25, 0.50, 1, 2, 5, 10
Dynamic (σ, ϑ)	$\sigma = 0.20$; $\vartheta \in \{0, 1, 10\}$
Temporal profile	Clustered, intermittent, shock–recovery
Changed-request set	One participant; local five; dispersed five
Comparison mechanism	Shortfall-cost benchmark; mismatch-only feedback $\beta_{\mathrm{ratio}} = 0.5$; ECF $\beta_{\mathrm{ECF}} = 0.25, 0.5, 1$; equal-total control
FA parameters	$\alpha_{\mathrm{FA}} = 0.20$ MW; $\epsilon_{\mathrm{FA}} = 10^{-6}$ MW; α_{FA} sensitivity: $0.05, 0.10, 0.20, 0.40, 0.80$ MW
Controlled-study decision tolerances	$\epsilon_x = 10^{-9}$ MW; $\tau_{\mathrm{rpl}} = \tau_E = 2 \times 10^{-10}$ MW
Horizon	16 intervals with exact tree frequencies
Sample size	1,620 trajectories; 25,920 matched interval comparisons, each AC-checked

Supplementary Table 4: Supporting comparisons and their role in the main findings. Each comparison uses the units and conditions stated in its cited section.

Question	Supporting comparison	Location
When can a larger request reduce delivery elsewhere?	Matched request choices, private payoff, and total delivery under five allocation rules.	S2; main Fig. 3
When does feedback help?	Completed-state replay, ceiling dynamics, equal-total requests, and controlled AC checks.	S3–S4, S8; main Figs. 4–5
What does the replay test add?	ECF against updates made without replay, both after every idle award and at a similar frequency; gain and record-error comparisons.	S11.2–S11.3, S12; main Fig. 7 and Table 1
Does the delivery contrast depend on placement or supply?	Five profile assignments, five PV capacities, nested sections, and representative LV models.	S13; Supplementary Figs. 4–5
Who gains and how do requests respond?	Participant delivery differences and an experience-based request-choice model.	S14

Throughout, $[\cdot]^+$ denotes the positive-part operator. A request-effect difference is changed minus forecast-based reference requests; a feedback difference is ECF minus the named control. The main paper generally reports power differences to one decimal place and network energy totals to two, retaining additional digits for small effects and numerical limits. Smaller energy differences are expressed in kWh rather than MWh, where 1 MWh = 1000 kWh. Tables retain additional precision, and differences and percentages are calculated before rounding.

Controlled results refer to the stated case or trajectory grid. Australian comparisons use paired network days. Section and model panels are reported at their own spatial units rather than pooled into one network-population estimate.

S1.1 Recurring Symbols

The following table collects notation used in several sections. Symbols used in only one derivation are defined where they first appear.

Supplementary Table 5: Symbols used across the supplementary derivations and simulations.

Symbol	Meaning
$i, j, t, b, \ell;\ \mathcal{N}, \mathcal{T}, \mathcal{B}, \mathcal{L}$	Participant indices i, j, interval index t, bus index b, branch index ℓ, and their sets.
$\mathcal{K}, \mathcal{G}, \nu(i)$	Changed-request set, distribution-network graph, and participant-to-bus map.
$\mathcal{R}, \mathcal{M};\ \Pi_{\mathcal{R}}, \mathcal{F}_{\mathcal{M}}$	Allocation rule and network model; allocation and feasible-set maps.
$\mathbf{z}, \boldsymbol{\xi}_{\text{net}}, \boldsymbol{\eta}_{\mathcal{R}}$	Exogenous network data, endogenous network variables, and rule parameters.
$r_i, r_i^{\text{cap}}, r_i^{\text{adm}}, x_i$	Submitted, capacity-limited, and admitted requests, and authorized export access (MW).
$C_i, a_i^{\text{fcst}}, a_i, a_i^{\text{prof}}, a_i^{\text{tel}}, y_i$	Connection bound, point forecast, available export, profile-derived available-export proxy, post-event replay record, and completed-interval delivery (MW).
$F_i(\cdot \mid s_i), \gamma_i;\ U_i, u_i, \vartheta$	Conditional cumulative distribution function (CDF) of available export and request multiplier; expected/realized payoff and normalized shortfall-cost parameter.
$\mathcal{N}_{\text{ua}}(t), g(t), \psi_i, \beta_{\text{ECF}}$	Unused-award group, indicator that all replay conditions hold, request ceiling, and ECF update gain.
$\mathbf{r}^{\text{rpl}}, \mathbf{x}^{\text{rpl}}, \mathbf{y}^{\text{rpl}}$	Completed-state replay request, access, and delivery vectors.
$X, Y, H, \Delta Y, M_x$	Total authorized access, total completed delivery, total unused award, signed delivery change, and allocation movement (MW before conversion to the reported kW values).
$\mathcal{V}_b, v_b, S_{\ell,e}, \bar{S}_\ell^{\text{syn}}$	AC voltage phasor and magnitude, branch-end complex power, and synthetic scenario rating.
WP, ER, EA, ME, FA	Weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation.
ECF, MR	Externality-conditioned counterfactual feedback and margin retention.

Some fixed-interval derivations suppress t after specifying the interval. The feedback and public-data sections retain interval or day/half-hour indices. The participant set is $\mathcal{N}$. Participant i submits request r_i and has connection bound $C_i > 0$. The private forecast is a_i^{fcst}. In the controlled study, a_i is simulated available active-power export and authorized access x_i gives delivery $y_i = \min\{x_i, a_i\}$. Its post-event replay record is exact, $a_i^{\text{tel}} = a_i$. The Australian study instead constructs the profile-derived proxy a_i^{prof} from scaled gross-PV generation and household consumption. It models control-dependent delivery as $y_i^c = \min\{x_i^c, a_i^{\text{prof}}\}$ and sets the primary replay record to $a_i^{\text{tel}} = a_i^{\text{prof}}$. Record sensitivities alter only a_i^{tel}. Thus public-case delivery is a model outcome rather than source-metered export. Sections S10–S12 define the construction and record sensitivities.

The connection bound first gives $r_i^{\text{cap}} = \min\{r_i, C_i\}$. Externality-conditioned counterfactual feedback (ECF) then applies the MW-valued ceiling $\psi_i \in [0, C_i]$. The request presented to the allocation rule is

$$r_i^{\text{adm}} = \min\{r_i^{\text{cap}}, \psi_i\}. \tag{S1}$$

These quantities are successive request stages: submission, application of the connection bound, and application of the ECF ceiling. Once ψ_i binds, a larger submitted request cannot increase r_i^{adm}. Fixed-interval derivations condition on the current ceiling. The gain β_{ECF} affects allocation through the ceiling trajectory across intervals. The behavioral model also includes a normalized shortfall cost, but that cost

enters request choice rather than the allocation map or ceiling update. The feeder graph and exogenous network data are $\mathcal{G}$ and $\mathbf{z}$, respectively.

For allocation rule $\mathcal{R}$ and network model $\mathcal{M}$, the allocation map is

$$\mathbf{x}^{\mathcal{R},\mathcal{M}} = \Pi_{\mathcal{R}}\Big(\mathbf{r}^{\mathrm{adm}}; \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}), \boldsymbol{\eta}_{\mathcal{R}}\Big). \tag{S2}$$

When no ECF ceiling binds, $\mathbf{r}^{\mathrm{adm}} = \mathbf{r}^{\mathrm{cap}}$. Otherwise, the same allocation map receives the ceiling-limited request in (S1). Rule parameters $\boldsymbol{\eta}_{\mathcal{R}}$ enter $\Pi_{\mathcal{R}}$. They do not alter the network-feasible set. The auxiliary margin-retention variant is examined in Section S9. In the no-feedback baseline, $\psi_i = C_i$.

S2 Request Choice and Delivery Externality

A request affects private payoff through the participant's own award, but affects total delivery through awards at every connection. The derivation below separates these paths using one conditional available-export model for both request choice and delivery. Continuous-regime derivatives give the local relationship; matched finite differences apply to the discrete output tree. The numerical classification additionally requires an increased unused award, as specified in Section S2.2.

Participant i observes private signal s_i and forms point forecast a_i^{fcst}. The conditional cumulative distribution function (CDF) $F_i(\cdot \mid s_i)$ describes uncertainty in available export. Because the connection bound C_i clips the distribution, the point forecast need not equal its clipped mean. Request, access, available export, and delivery obey

$$r_i(\gamma_i) = \gamma_i a_i^{\mathrm{fcst}}, \quad x_i(\gamma_i) = [\Pi_{\mathcal{R}}(\mathbf{r}^{\mathrm{adm}}(\gamma_i))]_i, \quad y_i(\gamma_i) = \min\{x_i(\gamma_i), a_i\}, \qquad \gamma_i \geq 1. \tag{S3}$$

Here $a_i \sim F_i(\cdot \mid s_i)$. The same distribution therefore supports the request decision and the later delivery realization. In a single-participant comparison, all other participants retain their reference requests. This construction generates internally consistent controlled requests; it is not an equilibrium model of participant behavior. Let $\vartheta \geq 0$ denote the modeled cost per unit of unused award, normalized by the value per unit of delivered export. The conditional expected payoff is

$$U_i(\gamma_i; \vartheta) = \bar{y}_i(\gamma_i) - \vartheta \bar{h}_i(\gamma_i) = \mathbb{E}\big[\min\{x_i(\gamma_i), a_i\} - \vartheta[x_i(\gamma_i) - a_i]^+ \mid s_i\big]. \tag{S4}$$

Here $\bar{y}_i(\gamma_i) = \mathbb{E}[\min\{x_i(\gamma_i), a_i\} \mid s_i]$ and $\bar{h}_i(\gamma_i) = \mathbb{E}[[x_i(\gamma_i) - a_i]^+ \mid s_i]$. For a realized outcome, the corresponding normalized payoff is $u_i(t) = y_i(t) - \vartheta[x_i(t) - y_i(t)]^+$. Thus U_i and u_i denote expected and realized quantities, respectively. The expectation is over available export after the request and award have been fixed. All derivatives below hold the conditional available-export CDFs and exogenous inputs fixed. Conditional arguments are suppressed when no ambiguity arises.

S2.1 Private-payoff and total-delivery effects

For a CDF F_i that is continuous at x_i,

$$\frac{\partial}{\partial x_i}\mathbb{E}[\min\{x_i, a_i\} \mid s_i] = 1 - F_i(x_i \mid s_i), \qquad \frac{\partial}{\partial x_i}\mathbb{E}[(x_i - a_i)^+ \mid s_i] = F_i(x_i \mid s_i). \tag{S5}$$

To see the first identity, write $\mathbb{E}[\min\{x_i, a_i\}] = \int_0^{x_i}[1 - F_i(u)]\,du$ for nonnegative available export. The second follows from $\mathbb{E}[(x_i - a_i)^+] = \int_0^{x_i} F_i(u)\,du$. Applying the chain rule with $J_{ji} = \partial x_j / \partial r_i$ proves the private gradient in the main paper:

$$\frac{\partial U_i}{\partial r_i} = J_{ii}\left[1 - F_i(x_i) - \vartheta F_i(x_i)\right]. \tag{S6}$$

If $J_{ii} > 0$ and the optimum is interior, its target award satisfies $F_i(x_i) = 1/(1+\vartheta)$. A zero shortfall cost therefore values the upper available-export tail, while a larger shortfall cost selects a lower award quantile.

For conditional expected aggregate delivery $\bar{Y} = \sum_j \mathbb{E}[\min\{x_j, a_j\} \mid s_j]$, the same chain rule gives

$$\frac{\partial \bar{Y}}{\partial r_i} = \sum_{j \in \mathcal{N}} J_{ji}[1 - F_j(x_j)]. \tag{S7}$$

Equating this derivative with the private-payoff derivative in (S6) gives

$$\vartheta_i^{\text{ext}} = -\frac{\sum_{j \neq i} J_{ji}[1 - F_j(x_j)]}{J_{ii} F_i(x_i)}, \qquad J_{ii} F_i(x_i) > 0. \tag{S8}$$

Equation (S8) gives the normalized shortfall cost that would locally align the participant's marginal payoff with the marginal change in total delivery. This value depends on the active allocation response, electrical location, competing requests, and available-export CDFs. It may change when the active set changes and need not be positive when access is slack. One fixed shortfall cost therefore cannot be expected to reproduce every local cross-participant delivery effect. Settlement and ECF are consequently evaluated as different interventions. The participant internalizes only its own payoff derivative. Under scarcity, some J_{ji} for $j \neq i$ are negative. Hence the local delivery externality condition is

$$\frac{\partial U_i}{\partial r_i} > 0, \qquad \frac{\partial \bar{Y}}{\partial r_i} < 0. \tag{S9}$$

The request-to-access Jacobian makes this local condition depend on the allocation rule. The condition describes opposing marginal incentives and delivery effects. The finite-case classification below also checks the unused-award increment. At a nondifferentiable boundary, paired finite differences replace the derivative.

For a finite request step, define the conditional expected outside-delivery increment and the participant's expected unused-award increment as

$$\begin{aligned} \Delta \bar{Y}_{\mathcal{N} \setminus \{i\}} &= \sum_{j \neq i} (\bar{y}_j^{\text{chg}} - \bar{y}_j^{\text{ref}}), \\ \Delta \bar{h}_i &= \bar{h}_i^{\text{chg}} - \bar{h}_i^{\text{ref}}. \end{aligned} \tag{S10}$$

The expected private-payoff increment is $\Delta \bar{y}_i - \vartheta \Delta \bar{h}_i$, whereas the expected aggregate delivered-export increment is $\Delta \bar{y}_i + \Delta \bar{Y}_{\mathcal{N} \setminus \{i\}}$. Equating them gives

$$\vartheta_{i,\Delta}^{\text{ext}} = -\frac{\Delta \bar{Y}_{\mathcal{N} \setminus \{i\}}}{\Delta \bar{h}_i}, \qquad \Delta \bar{h}_i > 0. \tag{S11}$$

The ratio is positive when delivery outside the changed participant falls and is defined only when the unused-award increment is positive. We evaluate (S11) for 0.1-step multiplier changes with three properties: own delivery rises, the own unused award rises, and total delivery falls. There are 4,608 such steps. The median ratio is 2.287, with an interquartile range of 1.102–5.718. Rule-specific medians range from 0.918 to 5.368. The variation shows why a single normalized shortfall cost cannot match the cross-participant delivery effect in every tested state. These ratios are state-specific diagnostics rather than estimates of an implemented tariff.

The controlled available-export tree is

$$a_i^{(k)} = \min\left\{C_i, (1 + \sigma \varsigma_k) a_i^{\text{fcst}}\right\}, \quad (\varsigma_k)_{k=1}^5 = (-2, -1, 0, 1, 2), \quad (\pi_k)_{k=1}^5 = \frac{1}{16}(1, 4, 6, 4, 1). \tag{S12}$$

Before clipping at C_i, the standardized support has zero mean and unit variance. The expectation in (S4) is evaluated exactly on this tree. The controlled request-choice set is the uniform grid $\Gamma = \{1.0, 1.1, \ldots, 2.0\}$. With the allocation rule, uncertainty scale, and reactive-power case held fixed, we write the smallest maximizer as $\gamma_i^\star(\vartheta, \sigma, \mathcal{R}, q_{\mathrm{pf}})$. The additional arguments enter through the available-export CDF and allocation response. The smallest maximizer resolves a payoff tie and gives a one-step best response while all other requests remain fixed.

The static request-choice calculations use the uncertainty grid

$$\sigma \in \{0, 0.05, 0.10, 0.20, 0.30\},$$

and the shortfall-cost grid

$$\vartheta \in \{0, 0.05, 0.10, 0.25, 0.50, 1, 2, 5, 10\}.$$

The controlled multi-interval experiment fixes $\sigma = 0.20$ and uses $\vartheta \in \{0, 1, 10\}$. These grids determine both the static request choices and the requests used in the controlled multi-interval study.

The deterministic special case is recovered when $a_i = a_i^{\mathrm{fcst}}$ almost surely. Relative to $\gamma_i = 1$, define

$$\Delta h_i^{\mathrm{det}}(\gamma_i) = [x_i(\gamma_i) - a_i^{\mathrm{fcst}}]^+ - [x_i(1) - a_i^{\mathrm{fcst}}]^+. \tag{S13}$$

Let $\Delta y_i(\gamma_i) = \min\{x_i(\gamma_i), a_i^{\mathrm{fcst}}\} - \min\{x_i(1), a_i^{\mathrm{fcst}}\}$. If both Δy_i and $\Delta h_i^{\mathrm{det}}$ are positive, the deterministic request is privately beneficial relative to the reference when

$$\vartheta < \vartheta_{\mathrm{BE}}(\gamma_i) = \frac{\Delta y_i(\gamma_i)}{\Delta h_i^{\mathrm{det}}(\gamma_i)}. \tag{S14}$$

The pairwise break-even ratio is a local comparison rather than a global optimality condition. The main request-choice analysis therefore uses the stochastic best response, while the deterministic request-grid comparison describes a conditional response.

S2.2 Allocation equality and matched externality classification

Allocation-equality metrics describe how authorized access is distributed. Request–access–delivery externality classification requires a higher modeled payoff and larger unused award for the changed-request group, together with lower total delivery. It does not measure total social welfare. The Jain-index change is reported separately and is relative to the reference allocation; a less negative change denotes a smaller decline, not an increase above that reference.

For a matched allocation pair, let $\Delta\mathbf{x} = \mathbf{x}^{\mathrm{chg}} - \mathbf{x}^{\mathrm{ref}}$. The half-$\ell_1$ allocation-movement metric is

$$M_x = \frac{1}{2}\|\Delta\mathbf{x}\|_1. \tag{S15}$$

The movement metric includes redistribution and any common contraction or expansion, so it is not a pure transfer between participants. Net access change and delivery loss are reported separately.

For a nonempty participant set of size n, the raw allocation Jain index is

$$J_{\mathrm{raw}}(\mathbf{x}) = \frac{(\sum_i x_i)^2}{n \sum_i x_i^2}. \tag{S16}$$

Let $\mathbf{C} = (C_i)_{i \in \mathcal{N}}$. For $\mathcal{N}_+ = \{i : C_i > 0\}$ and $n_+ = |\mathcal{N}_+|$, the capacity-normalized index is

$$J_{\mathrm{norm}}(\mathbf{x}, \mathbf{C}) = \frac{(\sum_{i \in \mathcal{N}_+} x_i/C_i)^2}{n_+ \sum_{i \in \mathcal{N}_+} (x_i/C_i)^2}. \tag{S17}$$

For a matched pair,

$$\Delta J_{\mathrm{norm}} = J_{\mathrm{norm}}(\mathbf{x}^{\mathrm{chg}}, \mathbf{C}) - J_{\mathrm{norm}}(\mathbf{x}^{\mathrm{ref}}, \mathbf{C}). \tag{S18}$$

Both indices require a nonempty participant set. The normalized index also requires at least one positive connection bound; participants with $C_i = 0$ are omitted from (S17). The raw index has no value at $\mathbf{x} = \mathbf{0}$, and the normalized index has no value when every retained ratio x_i/C_i is zero. All 30 controlled-study participants have $C_i > 0$, and every reported allocation is nonzero, so both indices are defined in every reported controlled case.

For a static matched allocation, define

$$\begin{aligned} \bar{Y}^{\mathrm{ref}} &= \sum_j \bar{y}_j^{\mathrm{ref}}, \qquad \bar{Y}^{\mathrm{chg}} = \sum_j \bar{y}_j^{\mathrm{chg}}, \\ \Delta\bar{Y} &= \bar{Y}^{\mathrm{chg}} - \bar{Y}^{\mathrm{ref}}. \end{aligned} \tag{S19}$$

For a nonempty changed-request set $\mathcal{K}$, let

$$\begin{aligned} \Delta U_{\mathcal{K}} &= \sum_{i\in\mathcal{K}} \left(U_i^{\mathrm{chg}} - U_i^{\mathrm{ref}} \right), \\ \Delta\bar{h}_{\mathcal{K}} &= \sum_{i\in\mathcal{K}} \left(\bar{h}_i^{\mathrm{chg}} - \bar{h}_i^{\mathrm{ref}} \right). \end{aligned} \tag{S20}$$

Using the externality-classification tolerance $\tau_E = 2 \times 10^{-10}$ MW, a static response case is a request–access–delivery externality only if

$$\Delta U_{\mathcal{K}} > \tau_E, \qquad \Delta\bar{h}_{\mathcal{K}} > \tau_E, \qquad \Delta\bar{Y} < -\tau_E. \tag{S21}$$

A productive static high request instead has $\Delta U_{\mathcal{K}} > \tau_E$ and $\Delta\bar{Y} \geq -\tau_E$. At $\vartheta = 0$, pooling all five uncertainty levels gives 1,500 cases: 1,073 satisfy the externality conditions (71.5%), 178 are productive high requests (11.9%), 220 retain the reference request (14.7%), and 29 larger requests meet neither high-request definition (1.9%). These four classes are mutually exclusive. Main Fig. 3(c) displays the same classification at $\sigma = 0.20$ rather than this pooled uncertainty grid.

The multi-interval experiment classifies realized intervals. Define

$$\begin{aligned} Y^{\mathrm{ref}}(t) &= \sum_j y_j^{\mathrm{ref}}(t), & H^{\mathrm{ref}}(t) &= \sum_j [x_j^{\mathrm{ref}}(t) - y_j^{\mathrm{ref}}(t)]^+, \\ Y^{\mathrm{chg}}(t) &= \sum_j y_j^{\mathrm{chg}}(t), & H^{\mathrm{chg}}(t) &= \sum_j [x_j^{\mathrm{chg}}(t) - y_j^{\mathrm{chg}}(t)]^+, \\ \Delta Y(t) &= Y^{\mathrm{chg}}(t) - Y^{\mathrm{ref}}(t). \end{aligned} \tag{S22}$$

Using the realized payoffs $u_i^{\mathrm{ref}}(t)$ and $u_i^{\mathrm{chg}}(t)$ defined above, set

$$\begin{aligned} \Delta u_{\mathcal{K}}(t) &= \sum_{i\in\mathcal{K}} [u_i^{\mathrm{chg}}(t) - u_i^{\mathrm{ref}}(t)], \\ \Delta H_{\mathcal{K}}(t) &= \sum_{i\in\mathcal{K}} \left([x_i^{\mathrm{chg}}(t) - y_i^{\mathrm{chg}}(t)]^+ - [x_i^{\mathrm{ref}}(t) - y_i^{\mathrm{ref}}(t)]^+ \right). \end{aligned} \tag{S23}$$

The interval is classified as a request–access–delivery externality only if

$$\Delta u_{\mathcal{K}}(t) > \tau_E, \qquad \Delta H_{\mathcal{K}}(t) > \tau_E, \qquad \Delta Y(t) < -\tau_E. \tag{S24}$$

A productive realized request increase instead has $\Delta u_{\mathcal{K}}(t) > \tau_E$ and $\Delta Y(t) \geq -\tau_E$. Both definitions require more than $\gamma_i > 1$. A fall in total delivery is not classified as an externality unless the changed-request group also benefits. For the signed realized-delivery decomposition, let $\Delta Y_{\mathcal{K}} = \sum_{i\in\mathcal{K}}(y_i^{\text{chg}} - y_i^{\text{ref}})$ and $\Delta Y_{\mathcal{N}\setminus\mathcal{K}} = \sum_{j\notin\mathcal{K}}(y_j^{\text{chg}} - y_j^{\text{ref}})$. Then

$$\Delta Y = \Delta Y_{\mathcal{K}} + \Delta Y_{\mathcal{N}\setminus\mathcal{K}}. \tag{S25}$$

A larger request can therefore be fully used or can increase total delivery. It becomes an externality under this definition only when the outside loss exceeds the delivery gain of the changed-request group.

This matched-pair label describes the effect of changing requests. It is distinct from the completed-state replay indicator $g(t)$ in Section S3, which determines whether ECF tightens a later request ceiling. The reported externality-interval share counts cases satisfying (S24); it is not the share with $g(t) = 1$. The former uses modeled group payoff relative to reference requests, whereas the latter uses completed operating records and a replay allocation.

Outside-loss implication. Let $\Delta\bar{Y}_{\mathcal{K}}$ and $\Delta\bar{Y}_{\mathcal{N}\setminus\mathcal{K}}$ denote the expected delivery changes inside and outside the changed set. Since $\Delta U_{\mathcal{K}} = \Delta\bar{Y}_{\mathcal{K}} - \vartheta\Delta\bar{h}_{\mathcal{K}}$ and $\vartheta \geq 0$, the static externality conditions imply

$$\Delta\bar{Y}_{\mathcal{K}} > \tau_E, \qquad \Delta\bar{Y}_{\mathcal{N}\setminus\mathcal{K}} = \Delta\bar{Y} - \Delta\bar{Y}_{\mathcal{K}} < -2\tau_E. \tag{S26}$$

Likewise, $\Delta u_{\mathcal{K}} = \Delta Y_{\mathcal{K}} - \vartheta\Delta H_{\mathcal{K}}$ and (S24) give

$$\Delta Y_{\mathcal{K}}(t) > \tau_E, \qquad \Delta Y_{\mathcal{N}\setminus\mathcal{K}}(t) < -2\tau_E. \tag{S27}$$

Thus, whenever the externality classification holds, the delivery gain of the changed set is accompanied by a larger delivery loss outside that set. This result supplies the formal basis for the classification used in the main paper; it does not determine whether ECF acts after a particular completed interval.

S3 Counterfactual Replay and Temporal Feedback

The replay tests whether a specified reallocation could have increased delivery elsewhere and in total during the completed interval. The ceiling update then determines how strongly that finding changes later requests. The distinction matters because output may recover after an unused award. This section defines the comparison, examines record error, and derives the ceiling's temporal properties before describing the controls and trajectories.

S3.1 Completed-state replay and its four conditions

The replay compares two allocations for the same completed interval. Only the admitted request vector changes. The network topology $\mathcal{G}$, exogenous network data $\mathbf{z}(t)$, participant set, allocation rule and parameters, and underlying available-export vector $\mathbf{a}(t)$ remain fixed. The reactive-power setting is part of $\mathbf{z}(t)$. For replayed access $\widetilde{\mathbf{x}}(t)$, delivery is $\widetilde{y}_i(t) = \min\{\widetilde{x}_i(t), a_i(t)\}$. The post-event record must describe this allocation-invariant $a_i(t)$. The resulting comparison asks what the same available export could have delivered under a different access allocation.

This interpretation applies when a different access award changes connection-point delivery but not the available export underlying the completed interval. The record may then come from a controller, inverter, or verifiable portfolio process. If access would also alter available export through inverter control, storage dispatch, flexible demand, or portfolio redispatch, that response must be included in the replay model. Section S10.2 states the corresponding operational information requirements.

For allocation $x_i(t)$, completed delivery $y_i(t)$, and the controlled-study unused-award tolerance $\epsilon_x = 10^{-9}$ MW, define

$$\mathcal{N}_{\mathrm{ua}}(t) = \{i : x_i(t) - y_i(t) > \epsilon_x\}. \tag{S28}$$

The completed-state replay request is

$$r_i^{\mathrm{rpl}}(t) = \begin{cases} \min\{r_i^{\mathrm{adm}}(t), y_i(t)\}, & i \in \mathcal{N}_{\mathrm{ua}}(t), \\ r_i^{\mathrm{adm}}(t), & i \notin \mathcal{N}_{\mathrm{ua}}(t). \end{cases} \tag{S29}$$

Let $\mathbf{x}^{\mathrm{rpl}}(t)$ be the allocation obtained from the same rule, network model, and rule parameters. Its delivery uses the specified available-export record,

$$y_i^{\mathrm{rpl}}(t) = \min\{x_i^{\mathrm{rpl}}(t), a_i^{\mathrm{tel}}(t)\}. \tag{S30}$$

The replay leaves the completed allocation unchanged and produces three comparison quantities:

$$\Delta X_{\mathrm{out}}^{+,\mathrm{rpl}}(t) = \sum_{j \notin \mathcal{N}_{\mathrm{ua}}(t)} [x_j^{\mathrm{rpl}}(t) - x_j(t)]^+, \tag{S31}$$

$$\Delta Y_{\mathrm{out}}^{+,\mathrm{rpl}}(t) = \sum_{j \notin \mathcal{N}_{\mathrm{ua}}(t)} [y_j^{\mathrm{rpl}}(t) - y_j(t)]^+, \tag{S32}$$

$$\Delta Y_{\mathrm{sys}}^{\mathrm{rpl}}(t) = \sum_j y_j^{\mathrm{rpl}}(t) - \sum_j y_j(t). \tag{S33}$$

The first two quantities sum positive participant-level changes outside the unused-award group. The third is the signed change in aggregate participant delivery; the subscript "sys" denotes the sum over all participating connections. It subtracts any delivery lost within the updated group as well as among other connections. Because replay requests of group members do not exceed their completed delivery, their replay delivery cannot exceed completed delivery. A positive aggregate change therefore implies a positive signed net change outside the group, while the positive-part quantity identifies outsiders whose delivery increases. The controlled study uses $\tau_{\mathrm{rpl}} = 2 \times 10^{-10}$ MW. The indicator $g(t)$ equals one only when the replay allocation is obtained, its alternating-current (AC) power flow passes the specified checks, and all three quantities defined immediately above exceed τ_{rpl}. Otherwise, $g(t) = 0$.

With an exact available-export record and delivery given by $y_i = \min\{x_i, a_i\}$, these quantities satisfy $\Delta Y_{\mathrm{sys}}^{\mathrm{rpl}} \le \Delta Y_{\mathrm{out}}^{+,\mathrm{rpl}} \le \Delta X_{\mathrm{out}}^{+,\mathrm{rpl}}$. The group cannot deliver more in the replay, and an outsider's delivery increase cannot exceed its award increase. With the same threshold for all three comparisons, the total-recovery condition therefore implies the other two. The outside quantities describe how access is released and used; they are not independent filters in the exact-record model. All three comparisons are retained when the available-export record is perturbed.

Thus $g(t) = 1$ qualifies the unused-award episode for a later update of every group member's ceiling. It establishes recoverable delivery for this group and this reallocation, rather than each member's marginal contribution. If the comparison fails, ECF retains the capacity target; it does not search all possible reallocations. The update changes later request ceilings, not individual payments or penalties.

The studies use different numerical thresholds, summarized in Table 6. They suppress numerical near-zero comparisons and do not represent instrument accuracy. In particular, the Australian 2-W threshold is distinct from the record-error radius δ_i. The error radii used in the simulations are prescribed perturbation levels; an operational use of the bound would require independent validation. Record-error and physical-feasibility tests are evaluated separately.

S3.2 Post-event record error and AC checks

Exact simulated record. When $a_i^{\mathrm{tel}}(t) = a_i(t)$, (S30) uses the exact simulated availability. The two delivery-recovery quantities, replay AC input, and $g(t)$ are then identical to the known-availability calculation.

Supplementary Table 6: Numerical thresholds used to classify unused awards and recovery. All entries are in MW. The economic externality classification is used only in the controlled study.

Test	Controlled study	Australian simulation
Unused award, ϵ_x	10^{-9}	2×10^{-6}
Replay recovery, τ_{rpl}	2×10^{-10}	2×10^{-6}
Economic classification, τ_E	2×10^{-10}	Not used

Conservative replay with bounded record error. The completed-state interpretation still requires availability to remain invariant under the replayed allocation. Now suppose the post-event record has a validated componentwise error radius $|a_i^{\mathrm{tel}}(t) - a_i(t)| \le \delta_i$ and metered delivery is exact. Set

$$a_i^{\mathrm{L}}(t) = \max\{y_i(t), a_i^{\mathrm{tel}}(t) - \delta_i\}, \qquad y_i^{\mathrm{rpl,L}}(t) = \min\{x_i^{\mathrm{rpl}}(t), a_i^{\mathrm{L}}(t)\}. \tag{S34}$$

Because $y_i(t) \le a_i(t)$ and $a_i^{\mathrm{tel}}(t) - \delta_i \le a_i(t)$,

$$y_i(t) \le a_i^{\mathrm{L}}(t) \le a_i(t), \qquad y_i^{\mathrm{rpl,L}}(t) \le \min\{x_i^{\mathrm{rpl}}(t), a_i(t)\}. \tag{S35}$$

The map $v \mapsto [v - y_i(t)]^+$ and summation are nondecreasing. Recovery outside the unused-award group and signed total recovery computed from $\mathbf{y}^{\mathrm{rpl,L}}(t)$ are therefore lower bounds on their values under the true completed-state availability. If both lower bounds exceed τ_{rpl}, the corresponding delivery-recovery conditions also hold for the true record. Allocation and network conditions are checked separately. The exact simulated record is recovered when $\delta_i = 0$.

The controlled-study access-award calculation checks the replayed allocation. Separate nonlinear radial AC calculations evaluate every distinct completed-delivery vector $\mathbf{y}(t)$ and replay-delivery vector $\mathbf{y}^{\mathrm{rpl}}(t)$ in the same network model. The reactive-power setting is either Q0 or 0.95-power-factor absorption (Q95). Each calculation must converge, satisfy the stated active- and reactive-power balance tolerances, keep bus voltages within 0.95–1.05 pu, and keep apparent power at both ends of every monitored branch within its synthetic scenario rating. A controlled trajectory passes only when its award, completed-delivery, and replay-delivery states all pass. Delivery-state feasibility is therefore evaluated directly rather than inferred from $\mathbf{y} \le \mathbf{x}$. Section S10.4 states the different customer-voltage and transformer checks used by the Australian models.

S3.3 Request-ceiling bounds, recovery, and timing

The update balances a lower ceiling after a qualifying replay with recovery toward the connection bound in other intervals. Both targets and current ceilings lie within that bound. Repeated averaging determines the rate of response; later delivery also depends on output and competing requests. The MW-valued target is

$$\psi_i^{\mathrm{tar}}(t) = \begin{cases} y_i(t), & g(t) = 1 \text{ and } i \in \mathcal{N}_{\mathrm{ua}}(t), \\ C_i, & \text{otherwise}, \end{cases} \tag{S36}$$

and ECF uses

$$\psi_i(t+1) = (1 - \beta_{\mathrm{ECF}})\psi_i(t) + \beta_{\mathrm{ECF}}\psi_i^{\mathrm{tar}}(t), \qquad \psi_i(1) = C_i. \tag{S37}$$

For $0 \le \beta_{\mathrm{ECF}} \le 1$ and $0 \le y_i(t) \le C_i$, the target lies in $[0, C_i]$, as does the initial ceiling. Each update in (S37) is a convex combination of the current ceiling and target, so induction gives

$$0 \le \psi_i(t) \le C_i \quad \text{for every } t. \tag{S38}$$

For a qualifying group member and $\beta_{\mathrm{ECF}} > 0$, the ceiling strictly decreases. Indeed, $y_i(t) < x_i(t) \le r_i^{\mathrm{adm}}(t) \le \psi_i(t)$, so

$$\psi_i(t+1) - \psi_i(t) = -\beta_{\mathrm{ECF}}[\psi_i(t) - y_i(t)] < 0. \tag{S39}$$

This is a statement about the request ceiling, not the next award or delivery, which also depend on other requests and available export.

If the target remains equal to a constant ψ_i^{fix} for $N \geq 1$ updates, repeated substitution gives

$$\psi_i(t+N) = (1-\beta_{\mathrm{ECF}})^N \psi_i(t) + \left[1-(1-\beta_{\mathrm{ECF}})^N\right]\psi_i^{\mathrm{fix}}. \tag{S40}$$

If none of N consecutive intervals meets all four replay conditions, the target stays at $\psi_i^{\mathrm{fix}} = C_i$ throughout those updates. The same expression then gives geometric recovery for $0 < \beta_{\mathrm{ECF}} \leq 1$. At unit gain, one interval with $g(t) = 0$ restores every ceiling to C_i at $t+1$. At $\beta_{\mathrm{ECF}} = 0$, the state remains unchanged. Full delivery gives $\mathcal{N}_{\mathrm{ua}}(t) = \varnothing$. An unused award also triggers no new tightening when the replay releases no access outside the group or recovers no completed-state delivery; existing ceilings then recover toward C_i. These cases show why an unused award need not lead to a later restriction.

The recursion uses no future information. The admitted request at t depends on $\psi_i(t)$, which contains completed records only through interval $t-1$. A submitted request cannot bypass the ceiling: $r_i^{\mathrm{adm}}(t) = \min\{r_i(t), C_i, \psi_i(t)\}$ cannot exceed $\psi_i(t)$ for any submitted request. When the target is fixed, a larger gain reaches it faster. In the complete loop, however, allocation and available export change the indicator and target. The fixed-target result therefore does not imply closed-loop convergence or monotonic delivery improvement. The primary comparison uses the intermediate gain $\beta_{\mathrm{ECF}} = 0.5$; gains of 0.25 and 1 test a slower update and full replacement by the latest target. Within each trajectory the gain is fixed: the replay indicator selects the target, not the gain. The gain comparisons therefore test preset response strengths rather than an adaptive gain-selection rule.

S3.4 Comparison controls

The shortfall-cost benchmark uses the request selected from (S4) but has no post-event request-ceiling state:

$$\psi_i^{\mathrm{cost}}(t) = C_i, \qquad r_i^{\mathrm{adm,cost}}(t) = r_i^{\mathrm{cap}}(t). \tag{S41}$$

The superscript cost denotes the shortfall-cost benchmark. Thus ϑ changes request choice before allocation, while completed delivery does not change a later admitted request. At $\vartheta = 0$, this comparison has neither a shortfall cost nor feedback.

Mismatch-only feedback tests a simpler update based on each participant's own delivery shortfall. It uses the delivery-to-award ratio below. The state starts at $\chi_i(1) \in [0,1]$ and uses $\beta_{\mathrm{ratio}} \in [0,1]$:

$$d_i(t) = \begin{cases} 1, & x_i(t) \leq \epsilon_x, \\ y_i(t)/x_i(t), & x_i(t) > \epsilon_x, \end{cases} \tag{S42}$$

$$\chi_i(t+1) = (1-\beta_{\mathrm{ratio}})\chi_i(t) + \beta_{\mathrm{ratio}} d_i(t). \tag{S43}$$

Its admitted request is

$$r_i^{\mathrm{adm,ratio}}(t+1) = \min\{r_i^{\mathrm{cap}}(t+1), \chi_i(t+1)C_i\}. \tag{S44}$$

Because $0 \leq y_i(t) \leq x_i(t)$, the ratio already lies in $[0,1]$. For an award above ϵ_x, a positive delivery shortfall gives $d_i(t) < 1$, setting a target below full admission. At a positive feedback gain, the state decreases when $d_i(t) < \chi_i(t)$ and increases when $d_i(t) > \chi_i(t)$; a shortfall therefore does not imply a further decrease from the previous state. At zero gain, the state is unchanged. If the award is at most ϵ_x or there is no delivery shortfall, $d_i(t) = 1$ and a positive-gain update moves the state toward full admission. Whether the admitted request changes also depends on which bound in (S44) is active. Mismatch-only feedback does not use the three replay-recovery quantities defined in Section S3.

S3.5 Controlled trajectories and physical coverage

Let $q_{\mathrm{pf}} \in \{\mathrm{Q0}, \mathrm{Q95}\}$ denote the reactive-power setting. Q0 sets participant reactive power to zero; Q95 specifies 0.95 power factor with reactive-power absorption. Equivalently, $\phi_{\mathrm{pf}} = 0$ for Q0 and $\phi_{\mathrm{pf}} = \arccos(0.95)$ for Q95. The controlled multi-interval test fixes $\sigma = 0.20$ and carries the multiplier that maximizes single-interval expected payoff in (S4) into the temporal loop:

$$r_i^\star(t) = \begin{cases} \gamma_i^\star(\vartheta, \sigma, \mathcal{R}, q_{\mathrm{pf}}) a_i^{\mathrm{fcst}}(t), & i \in \mathcal{K}, \\ a_i^{\mathrm{fcst}}(t), & i \notin \mathcal{K}, \end{cases} \qquad y_i(t) = \min\{x_i(t), a_i(t)\}. \tag{S45}$$

The realized $a_i(t)$ is a node of the same conditional tree used for request choice in (S12). The three 16-interval standardized sequences are

$$\begin{aligned} \boldsymbol{\varsigma}^{\mathrm{C}} &= (-2,-1,-1,-1,-1,0,0,0,0,0,0,1,1,1,1,2), \\ \boldsymbol{\varsigma}^{\mathrm{I}} &= (-2,0,1,-1,0,1,-1,0,1,-1,0,1,-1,0,0,2), \\ \boldsymbol{\varsigma}^{\mathrm{S}} &= (0,0,0,-1,-1,-2,-1,-1,0,0,0,1,1,1,1,2). \end{aligned} \tag{S46}$$

They represent clustered shortfall, intermittent shortfall, and shock–recovery. Each sequence contains the tree nodes with counts $(1,4,6,4,1)$. Participant i uses the cyclic phase offset $(7i) \bmod 16$. This preserves each participant's marginal distribution, but changing the base sequence also changes simultaneous node combinations across participants. These scenarios therefore vary temporal ordering and cross-participant coincidence together; they do not isolate persistence of a fixed set of whole-network states.

The multi-interval normalized shortfall-cost levels are $\vartheta \in \{0, 1, 10\}$. The changed set is either one participant, a topology-selected local set of five, or a dispersed set of five. In the multi-participant cases, each member uses its own $\gamma_i^\star$ from the corresponding single-participant request-choice calculation. All five changed requests are then applied simultaneously. This tests simultaneous requests chosen from individual response calculations; the participants do not jointly solve for an equilibrium. The multiplier is not re-optimized in response to later ECF ceilings. This keeps raw requests matched across controls and isolates their allocation responses. Experience-based adjustment of later requests is examined separately in the public-data model of Section S14.2; the controlled trajectories here do not include that response. Every ECF state starts at $\psi_i(1) = C_i$, while the mismatch-only feedback starts at $\chi_i(1) = 1$. All controls therefore use the same first allocation within each matched setting, before any completed delivery record is available.

Each of the six mechanisms contributes 270 trajectories of 16 intervals. The full design therefore contains 1,620 trajectories and 25,920 matched interval comparisons; each mechanism contributes 4,320 intervals. In every comparison, a changed participant's raw request is its selected multiplier times its forecast, and realized available export is a node of the same conditional tree. Paired mechanisms use identical raw requests, and delivery is $\min\{x_i, a_i\}$ componentwise.

In this controlled setting, the post-event record is simulation truth: $\mathbf{a}^{\mathrm{tel}}(t) = \mathbf{a}(t)$. The study evaluates 25,920 simulated deliveries. ECF contributes 12,960 replay attempts across its three gain settings, and all four replay conditions hold in 12,910 of them. Among these qualifying replays, median recovered delivery is 37.844 kW (interquartile range 25.606–57.527 kW), and 99.46% recover more than 5 kW. Every qualifying replay recovers more than 1.1 kW, well above the numerical comparison threshold. At the primary gain, 20 of 4,320 attempts recover no net delivery.

Both allocation states in every matched comparison pass the specified convergence, balance, voltage, and two-ended apparent-power checks under the same reactive-power setting. The delivery-state analysis covers 25,920 simulated-delivery occurrences and 12,960 replay-delivery occurrences. After duplicate states are removed within each reactive-power setting, 22,520 distinct states remain, and all satisfy the same criteria. Across the access-award and delivery-state calculations, the minimum voltage is 0.953772

Supplementary Table 7: Scale and AC power-flow coverage of the controlled study

Study component	Cases or states	Role or result
Static request-choice cases	13,500	Evaluated across the full static design
Static request–allocation pairs	3,600	All pass the specified AC checks
Behavior-linked interval inputs	4,320	Request and availability use the same tree
Multi-interval trajectories	1,620	Evaluated across all comparison mechanisms
Multi-interval allocation comparisons	25,920	Both states pass the specified AC checks
Simulated-delivery occurrences	25,920	All pass the specified AC checks
Replay-delivery occurrences	12,960	All pass the specified AC checks
Distinct reactive-power-tagged delivery states	22,520	All pass the specified AC checks

Supplementary Table 8: Delivery effects of the shortfall-cost benchmark and ECF relative to matched forecast-based reference requests.

ϑ	Control	$\bar{\gamma}^\star$	$\Delta Y_{\mathcal{K}}$ (kW)	$\Delta Y_{\mathcal{N}\setminus\mathcal{K}}$ (kW)	ΔY (kW)	Externality intervals (%)
0	Benchmark	1.513	15.948	−37.647	−21.699	50.1
0	Benchmark + ECF	1.513	12.665	−9.439	3.226	21.1
1	Benchmark	1.143	8.678	−9.357	−0.679	15.6
1	Benchmark + ECF	1.143	7.657	−0.612	7.045	8.9
10	Benchmark	1.045	0.071	−0.160	−0.090	0.0
10	Benchmark + ECF	1.045	1.104	5.887	6.991	1.3

per unit (pu), maximum branch loading is 90.91%, and no branch exceeds its synthetic scenario rating. The controlled lower-voltage limit is $V_{\min} = 0.95$ pu; hence the margin shown in Supplementary Fig. 3(a) is $10^3 \min_b[v_b - V_{\min}]$ in 10^{-3} pu.

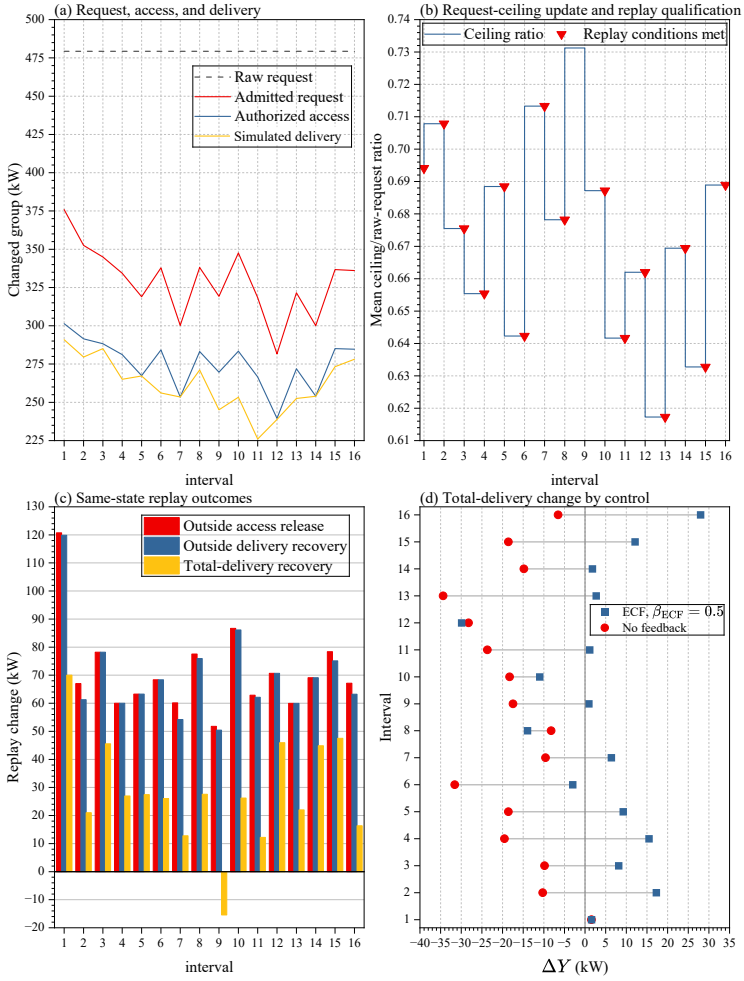
Auxiliary allocation-response coverage. The two static counts describe different grids. The 13,500 request-choice cases equal 30 participant locations × five allocation rules × two reactive-power settings × five uncertainty levels × nine shortfall-cost levels. The 3,600 matched request–allocation pairs equal 30 locations × five rules × two reactive-power settings × 12 response multipliers. The 12-point response surface includes the additional fixed multiplier 1.74. The behavioral choice grid instead contains the 11 values from 1.0 to 2.0 in increments of 0.1. Thus 1.74 is excluded from behavioral request selection, and the 3,600 pairs are not a subset count of the 13,500 cases.

Table 8 separates the delivery change of the changed-request group from the change for all other participants. A positive outside-group value denotes recovery; a negative value denotes loss. Here $\bar{\gamma}^\star$ is the mean selected multiplier across the changed-request cases. Increasing ϑ changes requests before allocation. ECF acts later and only after all replay conditions hold. At $\vartheta = 10$, the shortfall-cost benchmark has no classified externality interval. ECF still raises mean delivery, while the classification rate rises from zero to 1.3%. The replay test seeks recoverable delivery in a completed interval, not the minimum of this separate economic classification rate.

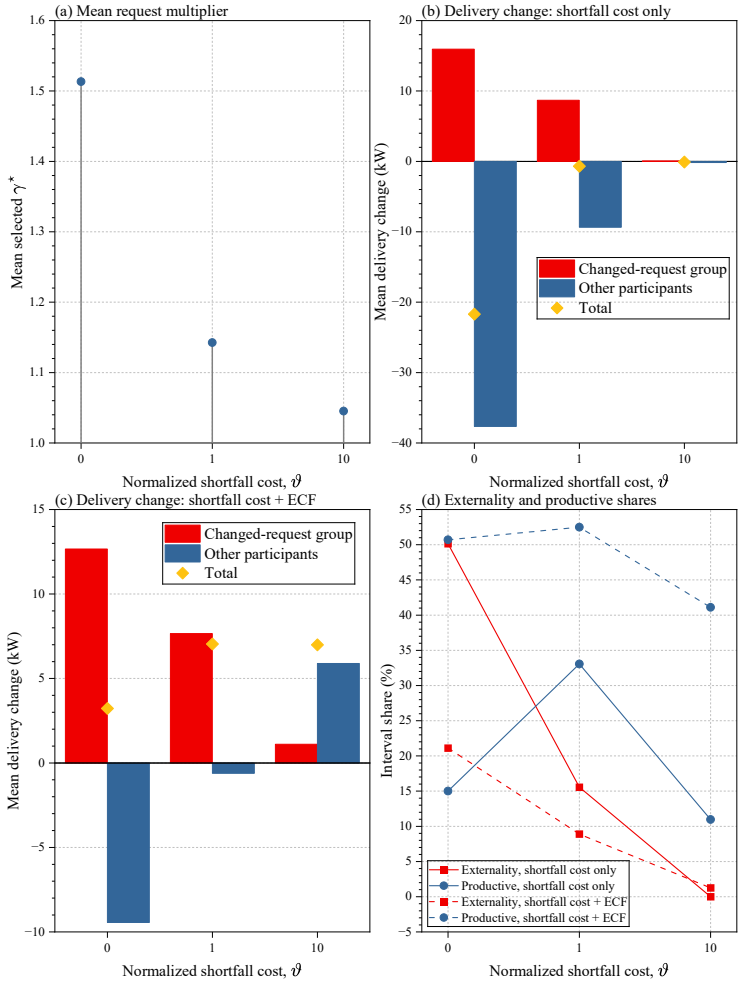
The comparison in Table 9 makes the two baselines explicit. Mismatch-only feedback recovers 15.197 kW relative to no feedback, and ECF recovers 24.925 kW against the same baseline. Their unrounded difference gives ECF's 9.727-kW advantage. The positive 3.226 kW in the first column instead compares ECF with forecast-based reference requests. These are different contrasts of the same trajectories.

Supplementary Table 9: Mean total-delivery comparisons at $\vartheta = 0$ and primary gain 0.5. Each value is in kW; the column heading identifies the subtracted baseline. Differences are calculated before rounding.

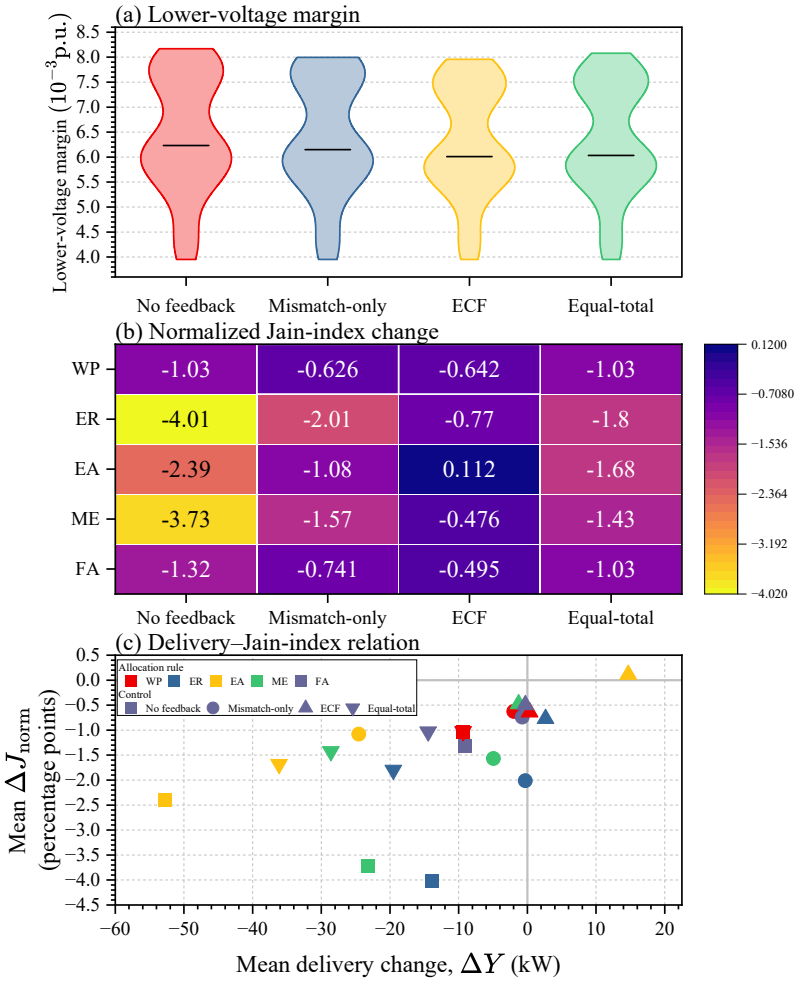
Control	Minus forecast-based reference requests	Minus no feedback
No feedback	−21.699	0.000
Mismatch-only feedback	−6.502	15.197
Equal-total control	−21.610	0.089
ECF	3.226	24.925



Supplementary Fig. 1: One 16-interval ECF trace for the local-five, shock–recovery, WP–Q0 case at $\vartheta = 0$. (a) Raw request, admitted request, awarded access, and simulated delivery for the changed-request group. (b) Mean of $\psi_i(t+1)/r_i(t)$ over the five changed-request participants, after each update. Red triangles mark intervals that meet the replay conditions. (c) Access released and delivery recovered outside the current unused-award group, with the signed change in total delivery. (d) Total-delivery change under ECF and no feedback, each relative to forecast-based reference requests. The controls coincide in interval 1 because no earlier delivery record exists. The local-five case changes five electrically nearby participants; shock–recovery is one 16-interval ordering of the same available-export-node frequencies. WP denotes weighted proportional allocation, and Q0 sets participant reactive power to zero.



Supplementary Fig. 2: How a modeled pre-allocation shortfall cost and post-delivery ECF affect requests and delivery. Results cover all profiles, spatial cases, allocation rules, and reactive-power settings. (a) Mean selected request multiplier. (b) Delivery changes relative to forecast-based reference requests with the shortfall cost alone. The bars separate the changed-request group and other participants; diamonds show their total. (c) The same changes with the shortfall cost and ECF at $\beta_{\mathrm{ECF}} = 0.5$. Positive blue bars denote recovery outside the changed-request group. (d) Externality and productive-high-request shares with and without ECF.



Supplementary Fig. 3: Network-limit and allocation-equality outcomes for the AC-validated controlled states. (a) Lower-voltage margin at $\vartheta = 0$, where the margin is the minimum bus-voltage magnitude minus the 0.95-pu lower limit. Violin width shows empirical density, and the black segment shows the median. (b) Mean change in the capacity-normalized Jain index, in percentage points, relative to the forecast-based reference allocation, by rule and control. Negative values indicate lower allocation equality than that reference. (c) Mean total-delivery change relative to forecast-based reference requests plotted against the corresponding Jain-index change. Colors identify allocation rules, and marker shapes identify no feedback, mismatch-only feedback, ECF, and the equal-total control. ECF and equal-total use $\beta_{\text{ECF}} = 0.5$.

S4 Equal-Total Request Control and Delivery Identity

Fixing the total request reduction leaves its distribution across participants as the distinction between ECF and the equal-total control. Electrical location and available export jointly determine the delivery effect of that distribution. The first identity below establishes that matching. The delivery identity then separates changes in total authorization from changes in its unused portion. Together, they explain why equal-total request reduction need not imply equal authorization or delivery.

S4.1 Matched request reduction and delivery decomposition

The paired comparison changes only the request transformation. The allocation rule, network model, interval, feeder data, and reactive-power setting remain fixed. Let $\mathbf{r}^{\mathrm{adm,ECF}}(t)$ be the admitted request produced by the ECF trajectory used for the comparison. The reported primary control is matched to $\beta_{\mathrm{ECF}} = 0.5$. With $r_i^{\mathrm{cap}}(t) = \min\{r_i(t), C_i\}$, define

$$R_{\mathrm{cap}}(t) = \sum_i r_i^{\mathrm{cap}}(t), \qquad \Delta R_{\mathrm{ECF}}(t) = \sum_i \left[r_i^{\mathrm{cap}}(t) - r_i^{\mathrm{adm,ECF}}(t) \right]. \tag{S47}$$

For $R_{\mathrm{cap}}(t) > 0$, the equal-total control uses the following scaling factor and proportional request:

$$\begin{aligned} \upsilon(t) &= 1 - \frac{\Delta R_{\mathrm{ECF}}(t)}{R_{\mathrm{cap}}(t)}, \\ r_i^{\mathrm{adm,uni}}(t) &= \upsilon(t) r_i^{\mathrm{cap}}(t). \end{aligned} \tag{S48}$$

The superscript uni denotes the uniform scaling used by the equal-total control. Each admitted ECF request lies between zero and its capacity-limited request. Hence the removed total lies between zero and $R_{\mathrm{cap}}(t)$, which gives $0 \le \upsilon(t) \le 1$. If $R_{\mathrm{cap}}(t) = 0$, set $\upsilon(t) = 1$; both the capacity-limited and equal-total request vectors are then zero. For a positive total request, direct substitution gives

$$\sum_i \left[r_i^{\mathrm{cap}}(t) - r_i^{\mathrm{adm,uni}}(t) \right] = \Delta R_{\mathrm{ECF}}(t). \tag{S49}$$

Thus ECF and the equal-total control remove the same request total but generally remove it at different connections. The control obtains its reduction total from ECF and distributes it proportionally. It therefore tests the location of the reduction, rather than an independent operating policy or the contribution of individual replay conditions.

The mismatch-only comparison in Section S3 evaluates two complete feedback rules. Mismatch-only feedback uses each participant's delivery-to-award ratio. ECF uses a delivery target when the completed-state comparison qualifies the unused-award group. The information and update formulas both differ, so their delivery contrast concerns the complete rules. Section S11.2 separates the role of qualification by holding the target formula, update weight, and recovery law fixed in a separate Australian comparison.

For the following interval-level identity, time arguments are suppressed. Because $0 \le y_i \le x_i$ in both studies' delivery models, $Y = \sum_i y_i = X - H$, where $X = \sum_i x_i$ and $H = \sum_i (x_i - y_i)$. Consequently, the paired delivery advantage of ECF over the equal-total control decomposes exactly as

$$Y^{\mathrm{ECF}} - Y^{\mathrm{uni}} = \left(X^{\mathrm{ECF}} - X^{\mathrm{uni}} \right) + \left(H^{\mathrm{uni}} - H^{\mathrm{ECF}} \right). \tag{S50}$$

More generally, compare two allocations under the same available export, and write each difference as the second allocation minus the first. Then

$$\Delta Y = \Delta X - \Delta H, \qquad \Delta Y > 0 \iff \Delta X > \Delta H. \tag{S51}$$

The condition $\Delta X > \Delta H$ covers both signs of the authorization change. If authorization falls, its unused portion must fall by more. If authorization rises, delivery also rises whenever any increase in the unused portion is smaller. Multiplying by interval duration and summing gives the energy identity used in the public-data results. It decomposes an outcome without assigning independent causal effects to its two terms.

A feasible redistribution that increases delivery. Fix one interval and its available-export vector. Suppose participant i has $x_i > a_i$, while participant j has $x_j < a_j$. Reduce i's award by a positive amount d_x and increase j's award by a positive amount e_x, with

$$0 < d_x \le x_i - a_i, \qquad 0 < e_x \le a_j - x_j. \tag{S52}$$

Suppose the new allocation satisfies the applicable request bounds and network constraints, and all other awards remain unchanged. Since delivery is $\min\{x_i, a_i\}$, the reduction leaves i's delivery unchanged, whereas j delivers e_x more. Consequently,

$$\Delta Y = e_x, \qquad \Delta X = e_x - d_x, \qquad \Delta H = -d_x. \tag{S53}$$

If $d_x = e_x$, delivery increases at unchanged total authorization. If $d_x > e_x$, delivery increases while both total and unused authorization fall. If $e_x > d_x$, delivery increases together with total authorization while unused authorization still falls. These cases give different signs of ΔX under the same delivery identity. Network feasibility is essential: access at different connections cannot generally be transferred without regard to electrical location. ECF changes requests rather than directly imposing this transfer, so the argument gives a sufficient condition on the resulting awards, not an automatic property of ECF.

Finite changes and request sensitivity. For any two fixed award vectors $\mathbf{x}^0$ and $\mathbf{x}^1$ under the same available-export vector, the signed-integral form is

$$Y^1 - Y^0 = \sum_i \int_{x_i^0}^{x_i^1} \mathbb{I}\{u < a_i\}\, \mathrm{d}u. \tag{S54}$$

Reversing the integration limits accounts for an award reduction. Under the same conditional available-export distributions, expected delivery instead satisfies

$$\bar{Y}(\mathbf{x}^1) - \bar{Y}(\mathbf{x}^0) = \sum_i \int_{x_i^0}^{x_i^1} [1 - F_i(u)]\, \mathrm{d}u. \tag{S55}$$

For this identity, both award vectors are fixed and F_i is the conditional cumulative distribution specified in Section S2. The relation requires no independence between participants' available exports. Together with the request-to-award derivatives in Section S2, it explains why the delivery effect depends on both the electrical response of allocation and the available export at the affected connections. A valid completed-state replay does not ensure that these conditions persist into the next interval.

Equal-total requests under proportional allocation. For the Australian rule, let c denote a comparison control and $R^c = \sum_i r_i^{\mathrm{adm},c}$ in a fixed half-hour. Because $x_i^c = \rho^c r_i^{\mathrm{adm},c}$, total authorization is $X^c = \rho^c R^c$. ECF and its equal-total control share $R^c = R$, giving

$$Y^{\mathrm{ECF}} - Y^{\mathrm{uni}} = R(\rho^{\mathrm{ECF}} - \rho^{\mathrm{uni}}) - (H^{\mathrm{ECF}} - H^{\mathrm{uni}}). \tag{S56}$$

The identity holds for the feasible factor ρ^c returned by the specified allocation search, whether or not that factor is globally maximal. Different spatial request distributions can yield different feasible factors, so equal-total request reduction need not produce equal-total authorization. If $R = 0$, both allocations and deliveries are zero.

Uniform reduction also preserves the relative requests of the no-feedback allocation. Let the common reduction factor be $\upsilon > 0$ as in (S48). If the no-feedback allocation factor is no larger than υ, dividing that factor by υ gives an admissible factor for the uniformly reduced request and reproduces the same award at every connection. Thus the no-feedback award remains a feasible candidate. If both searches

return that same point on the binding network constraint, uniform reduction leaves awards and delivery unchanged. This conditional scaling property explains how the two controls can have nearly equal delivery even though their admitted-request totals differ. It does not assert that the numerical search always returns that point or a global maximum. ECF generally changes relative requests, so this cancellation does not apply to it. The near-equality of the two control totals in Table 12 is consistent with the scaling property.

Across the primary 80-day simulation, equal-total delivery exceeds no-feedback delivery by only 0.235 kWh. The sum of absolute half-hour delivery differences is 2.690 kWh, so their close totals do not conceal large offsetting delivery changes. Requests are uniformly reduced in 969 half-hours. In 967 of them, the no-feedback allocation factor satisfies the condition above, making its award vector a feasible candidate after uniform reduction. This explains the nearly coincident deliveries while retaining the interval-level differences.

Section S3.5 reports the controlled trajectories and their replay-recovery distribution. In each interval at the primary gain, the equal-total control matches ECF's aggregate request reduction, separating its amount from its distribution across participants.

Part II. Controlled network experiments

S5 One-Parameter Allocation Rules

Weighted proportional (WP), equal reduction (ER), and equal access (EA) each select an allocation from a one-parameter family. In the controlled study, the common scale, reduction, or cap is chosen against the full conic proxy domain in Section S8. The simpler expressions below explain the three rules under a nonnegative linear allowance or one equal-weight aggregate allowance. They are analytical special cases, not alternative network models. ECF changes the admitted request without changing these rules. The primary experiments use the full allowance, corresponding to $\kappa = 0$ below. For each positive linear allowance B_m, write $\widetilde{B}_m = (1-\kappa)B_m$ to include the auxiliary margin-retention comparison. Section S9 defines that comparison in the full conic domain. These rules operate on the admitted request r_i^{adm} in (S1). For the linear-allowance WP special case, assume $h_{m,i} \geq 0$ and define

$$D_m = \sum_{j \in \mathcal{N}} h_{m,j} r_j^{\mathrm{adm}}, \qquad \tilde{\rho} = \min\left\{1, \min_{m: D_m > 0} \frac{\widetilde{B}_m}{D_m}\right\}, \qquad x_i^{\mathrm{WP},\mathcal{M}}(\mathbf{r}^{\mathrm{adm}}; \kappa) = \tilde{\rho} r_i^{\mathrm{adm}}. \tag{S57}$$

The admitted requests themselves are the proportional weights; there is no additional participant-weight parameter in this WP rule. Rows with $D_m = 0$ do not restrict $\tilde{\rho}$; if all rows have zero weighted request, set $\tilde{\rho} = 1$. Nonnegative coefficients and requests then give

$$\sum_i h_{m,i} x_i^{\mathrm{WP},\mathcal{M}}(\mathbf{r}^{\mathrm{adm}}; \kappa) \leq \widetilde{B}_m$$

for every monitored allowance m.

For the single equal-weight aggregate allowance $\sum_i x_i \leq B$, let $B > 0$ denote the original allowance and $\tilde{B} = (1-\kappa)B$. For ER, suppose this allowance binds and the nonempty set $\mathcal{A}_+ = \{i : r_i^{\mathrm{adm}} > \tilde{\lambda}\}$ remains fixed. Then

$$\tilde{\lambda} = \frac{\sum_{i \in \mathcal{A}_+} r_i^{\mathrm{adm}} - \tilde{B}}{|\mathcal{A}_+|}, \qquad x_i^{\mathrm{ER},\mathcal{M}}(\mathbf{r}^{\mathrm{adm}}; \kappa) = \max\{0, r_i^{\mathrm{adm}} - \tilde{\lambda}\}. \tag{S58}$$

Hence $\sum_i x_i^{\mathrm{ER},\mathcal{M}}(\mathbf{r}^{\mathrm{adm}}; \kappa) = \tilde{B}$ in that regime. For EA, let the nonempty capped set $\mathcal{C}$ and the uncapped set $\mathcal{U} = \mathcal{N} \setminus \mathcal{C}$ remain fixed. When the same aggregate allowance binds,

$$\tilde{\mu} = \frac{\tilde{B} - \sum_{i \in \mathcal{U}} r_i^{\mathrm{adm}}}{|\mathcal{C}|}, \qquad x_i^{\mathrm{EA},\mathcal{M}}(\mathbf{r}^{\mathrm{adm}}; \kappa) = \min\{r_i^{\mathrm{adm}}, \tilde{\mu}\}. \tag{S59}$$

This construction gives $\sum_i x_i^{\mathrm{EA},\mathcal{M}}(\mathbf{r}^{\mathrm{adm}};\kappa) = \tilde{B}$. When all relevant allowances are slack, the allocation equals the admitted request. These expressions apply only while the stated allowance row and membership sets remain fixed. The controlled study selects each common scale, reduction, or cap against the full conic proxy. If a binding constraint or set membership changes, the allocation is recomputed over that domain rather than extrapolated from the formula for one aggregate allowance. The three rules thus explain how a request change can redistribute authorized access, while the separate AC calculation determines whether the resulting state is physically feasible.

S6 Maximum-Export and Fairness-Aware Allocation

S6.1 Deterministic maximum-export selection

Maximum export (ME) maximizes total authorized access, which can leave several equally good allocation vectors. A deterministic second stage resolves this ambiguity so that arbitrary tie-breaking does not enter the matched request comparison. The construction below bounds the loss of the primary objective and proves uniqueness of the selected allocation in the convex proxy. The objective remains authorized access rather than realized delivery.

Here $\mathbf{r}^{\mathrm{adm}}$ is the request presented to the optimizer after ECF or a comparison control has acted. It already satisfies the capacity and feedback ceilings in (S1). Define the allocation-feasible projection

$$\mathcal{X}_{\mathcal{M}}(\mathbf{r}^{\mathrm{adm}}) = \left\{\mathbf{x} :\ 0 \le x_i \le r_i^{\mathrm{adm}}, \quad \exists \boldsymbol{\xi}_{\mathrm{net}} \text{ such that } (\mathbf{x}, \boldsymbol{\xi}_{\mathrm{net}}) \in \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z})\right\}. \tag{S60}$$

The MR case follows by replacing $\mathcal{F}_{\mathcal{M}}$ with $\mathcal{F}_{\mathcal{M}}^{\mathrm{MR}}$. Let $\mathbf{1}_n$ denote the all-ones vector of dimension $|\mathcal{N}|$. Stage 1 determines

$$X^\star = \max_{\mathbf{x} \in \mathcal{X}_{\mathcal{M}}(\mathbf{r}^{\mathrm{adm}})} \mathbf{1}_n^{\mathsf{T}} \mathbf{x}. \tag{S61}$$

Let $\mathbf{x}^{(1)}$ be a feasible Stage 1 allocation and set $X_1^{\mathrm{feas}} = \mathbf{1}_n^{\mathsf{T}} \mathbf{x}^{(1)}$. Let X_1^{ub} be a valid global dual upper bound for the same problem. The two bounds satisfy

$$X_1^{\mathrm{feas}} \le X^\star \le X_1^{\mathrm{ub}}, \qquad X_1^{\mathrm{ub}} - X_1^{\mathrm{feas}} \le \tau_H/4. \tag{S62}$$

The total admitted request and hierarchy tolerance are

$$R_{\mathrm{adm}} = \sum_i r_i^{\mathrm{adm}}, \qquad \tau_H = \tau_H^{\mathrm{abs}} + \tau_H^{\mathrm{rel}} R_{\mathrm{adm}}, \qquad \tau_H^{\mathrm{abs}} = 2 \times 10^{-7} \text{ MW}, \quad \tau_H^{\mathrm{rel}} = 2 \times 10^{-8}. \tag{S63}$$

The superscripts "abs" and "rel" denote the absolute and relative components of the hierarchy tolerance. For $R_{\mathrm{adm}} > 0$, Stage 2 selects

$$\begin{aligned} \mathbf{x}^{\mathrm{ME},\mathcal{M}} = \arg \min_{\mathbf{x} \in \mathcal{X}_{\mathcal{M}}(\mathbf{r}^{\mathrm{adm}})} \quad & \bar{\mathbf{w}}^{\mathsf{T}} \mathbf{x} + \frac{\eta_{\mathrm{ME}}}{2R_{\mathrm{adm}}} \|\mathbf{x}\|_2^2 \\ \text{s.t.} \quad & \mathbf{1}_n^{\mathsf{T}} \mathbf{x} \ge X_1^{\mathrm{ub}} - \tau_H, \end{aligned} \tag{S64}$$

with

$$\bar{w}_i = \frac{\mathrm{ord}(i) - (n+1)/2}{n-1}, \quad \mathrm{ord} : \mathcal{N} \to \{1, \dots, n\}, \quad n = |\mathcal{N}| \ge 2, \quad \eta_{\mathrm{ME}} = 10^{-2}, \tag{S65}$$

where $\mathrm{ord}(i)$ is participant i's position in a fixed ordering. The coefficient η_{ME} is an ME rule parameter. It does not alter the network-feasible set $\mathcal{F}_{\mathcal{M}}$.

Proposition 1 (Deterministic maximum-export selection)**.** *Assume $R_{\mathrm{adm}} > 0$ and that $\mathcal{X}_{\mathcal{M}}(\mathbf{r}^{\mathrm{adm}})$ is nonempty, compact, and convex. If the Stage 1 certificate satisfies* (S62)*, then the Stage 2 feasible set is nonempty, its allocation-vector minimizer is unique, and*

$$0 \le X^\star - \mathbf{1}_n^{\mathsf{T}} \mathbf{x}^{\mathrm{ME},\mathcal{M}} \le \tau_H. \tag{S66}$$

Proof. The certificate gives $X_1^{\text{feas}} \geq X_1^{\text{ub}} - \tau_H/4 \geq X_1^{\text{ub}} - \tau_H$, so the Stage 1 allocation $\mathbf{x}^{(1)}$ satisfies the Stage 2 floor. The Stage 2 feasible set is therefore nonempty; it is also compact and convex. Its objective has Hessian $(\eta_{\text{ME}}/R_{\text{adm}})\mathbf{I}_n \succ \mathbf{0}$, so its allocation-vector minimizer is unique (the auxiliary network variables need not be unique). Feasibility in $\mathcal{X}_{\mathcal{M}}$ gives $\mathbf{1}_n^{\mathsf{T}}\mathbf{x}^{\text{ME},\mathcal{M}} \leq X^\star$, while the Stage 2 floor and $X^\star \leq X_1^{\text{ub}}$ give $X^\star - \mathbf{1}_n^{\mathsf{T}}\mathbf{x}^{\text{ME},\mathcal{M}} \leq \tau_H$. □

The proposition concerns the convex allocation proxy; nonlinear AC equations are evaluated after allocation. If $R_{\text{adm}} = 0$ and the zero-export baseline is feasible, the request bounds force $\mathbf{x} = \mathbf{0}$, so Stage 2 is unnecessary.

S6.2 Fairness-aware allocation

FA balances total authorized access against differences in the fractions of requests that are granted. For the same admitted requests, exogenous network data and reactive-power setting, first compute the Stage 2 ME allocation above. Its mean request-normalized fraction is

$$\bar{f}_{\text{ME}}^{\mathcal{M}} = \frac{1}{n}\sum_{i\in\mathcal{N}} \frac{x_i^{\text{ME},\mathcal{M}}}{r_i^{\text{adm}} + \epsilon_{\text{FA}}}. \tag{S67}$$

FA then solves

$$\mathbf{x}^{\text{FA},\mathcal{M}} = \arg\max_{\mathbf{x}\in\mathcal{X}_{\mathcal{M}}(\mathbf{r}^{\text{adm}})} \left\{ \sum_i x_i - \alpha_{\text{FA}} \sum_i \left(\frac{x_i}{r_i^{\text{adm}} + \epsilon_{\text{FA}}} - \bar{f}_{\text{ME}}^{\mathcal{M}} \right)^2 \right\}. \tag{S68}$$

The positive offset ϵ_{FA} prevents division by zero, and $\alpha_{\text{FA}} > 0$ weights the quadratic allocation-equality term. Both parameters have units of MW; the fractions are dimensionless. For fixed admitted requests, the objective is strictly concave in $\mathbf{x}$, giving a unique allocation on the nonempty compact convex domain. The reference fraction is recomputed when the requests change. FA uses the ME solution to set that reference, but does not impose the ME objective floor, so total authorization can fall as the equality term receives more weight. The controlled study uses $n = 30$ and the values in Table 3.

S7 Karush–Kuhn–Tucker Chain Rule for Request Sensitivity

The local response includes more than the direct request bound. Changing a request can also change the ME hierarchy quantities or the FA reference allocation. The chain rule below includes these dependencies within one smooth regime. At a kink or active-set change, recomputing both allocations gives the finite-difference response used by the experiments.

For fixed ceilings $\{\psi_i\}_{i\in\mathcal{N}}$, the submitted request affects both the allocation objective and its upper bound through the same admitted request. Let $\mathbf{e}_{i_0}$ be the i_0th coordinate vector. Away from the two kinks in (S1),

$$\frac{\partial \mathbf{r}^{\text{adm}}}{\partial r_{i_0}} = \mathbf{e}_{i_0}\, \mathbb{I}\{r_{i_0} < C_{i_0}\}\, \mathbb{I}\{r_{i_0}^{\text{cap}} < \psi_{i_0}\}. \tag{S69}$$

Here $\mathbb{I}\{\cdot\}$ is the indicator function. The derivative is zero after either the connection bound or ECF ceiling binds. A larger submitted request can no longer change the admitted request. Submitting an even larger value therefore cannot circumvent either bound.

Let $X_{\text{min}}^{\text{ME}} = X_1^{\text{ub}} - \tau_H$ denote the objective floor used by ME Stage 2. FA uses the request-dependent ME reference in (S67). The fixed rule parameters $\boldsymbol{\eta}_{\mathcal{R}}$ are distinct from the context quantities recomputed whenever the admitted request changes. For $\mathcal{R} \in \{\text{ME}, \text{FA}\}$, these quantities are

$$\boldsymbol{\theta}_{\text{ME}} = (R_{\text{adm}}, X_{\text{min}}^{\text{ME}}), \qquad \boldsymbol{\theta}_{\text{FA}} = (\bar{f}_{\text{ME}}^{\mathcal{M}}). \tag{S70}$$

The ME hierarchy enters the FA response through the derivative of $\bar{f}^{\mathcal{M}}_{\mathrm{ME}}$; R_{adm} and $X^{\mathrm{ME}}_{\min}$ do not become additional FA constraints. Let $\mathbf{p} = (\mathbf{x}, \boldsymbol{\xi}_{\mathrm{net}})$ collect the primal variables and $\boldsymbol{\omega}$ the active Karush–Kuhn–Tucker (KKT) multipliers. The KKT residual system is

$$\mathbf{F}^{\mathcal{R},\mathcal{M}}_{\mathrm{KKT}}(\mathbf{p}, \boldsymbol{\omega}; \mathbf{r}^{\mathrm{adm}}, \mathbf{z}, \kappa, \boldsymbol{\eta}_{\mathcal{R}}, \boldsymbol{\theta}_{\mathcal{R}}) = \mathbf{0}. \tag{S71}$$

For local sensitivity, $X^{\mathrm{ub}}_1(\mathbf{r}^{\mathrm{adm}})$ denotes the valid upper-bound certificate under a locally fixed, differentiable selection of Stage 1 primal and dual solutions. The derivative below is conditional on that selection and a strongly regular regime: the active cone face is unchanged, the reduced KKT Jacobian is nonsingular, and the ME bound quantities are differentiable. These conditions are not assumed at every solution returned by a conic solver.

$$\frac{d}{dr_{i_0}}\begin{bmatrix}\mathbf{p}\\ \boldsymbol{\omega}\end{bmatrix} = -\left(\frac{\partial \mathbf{F}_{\mathrm{KKT}}}{\partial(\mathbf{p},\boldsymbol{\omega})}\right)^{-1}\left(\frac{\partial \mathbf{F}_{\mathrm{KKT}}}{\partial \mathbf{r}^{\mathrm{adm}}}\frac{\partial \mathbf{r}^{\mathrm{adm}}}{\partial r_{i_0}} + \frac{\partial \mathbf{F}_{\mathrm{KKT}}}{\partial \boldsymbol{\theta}_{\mathcal{R}}}\frac{d\boldsymbol{\theta}_{\mathcal{R}}}{dr_{i_0}}\right). \tag{S72}$$

The ME hierarchy contributes

$$\frac{dR_{\mathrm{adm}}}{dr_{i_0}} = \mathbf{1}_n^{\mathsf{T}}\frac{\partial \mathbf{r}^{\mathrm{adm}}}{\partial r_{i_0}}, \qquad \frac{d\tau_H}{dr_{i_0}} = \tau_H^{\mathrm{rel}}\frac{dR_{\mathrm{adm}}}{dr_{i_0}}, \qquad \frac{dX^{\mathrm{ME}}_{\min}}{dr_{i_0}} = \frac{dX^{\mathrm{ub}}_1}{dr_{i_0}} - \tau_H^{\mathrm{rel}}\frac{dR_{\mathrm{adm}}}{dr_{i_0}}. \tag{S73}$$

For FA, the corresponding reference has the additional derivative

$$\frac{d\bar{f}^{\mathcal{M}}_{\mathrm{ME}}}{dr_{i_0}} = \frac{1}{n}\sum_{i\in\mathcal{N}}\left[\frac{1}{r_i^{\mathrm{adm}} + \epsilon_{\mathrm{FA}}}\frac{dx_i^{\mathrm{ME},\mathcal{M}}}{dr_{i_0}} - \frac{x_i^{\mathrm{ME},\mathcal{M}}}{(r_i^{\mathrm{adm}} + \epsilon_{\mathrm{FA}})^2}\frac{\partial r_i^{\mathrm{adm}}}{\partial r_{i_0}}\right]. \tag{S74}$$

The $\mathbf{x}$ block of (S72) is the local request-to-access response $\mathbf{J}_x$, whose entries are the J_{ji} used in main Methods. Equations (S72)– (S74) include both the admitted-request change and the request dependence of the ME or FA context quantities $\boldsymbol{\theta}_{\mathcal{R}}$. These derivatives apply only within the stated smooth regime. All reported finite request changes are evaluated by solving both allocations and taking their paired difference.

S8 Controlled Medium-Voltage Allocation Proxy and AC Verification

This section separates allocation from physical verification. All five controlled allocation rules are evaluated against the conic proxy defined below. WP, ER, and EA restrict the allocation to their respective one-parameter families, whereas ME and FA optimize directly over the full proxy domain. Each distinct access-award, simulated-delivery, and replay-delivery state is then evaluated by nonlinear radial AC power flow.

The cited 141-bus feeder[8] does not supply the branch thermal ratings used in this controlled experiment. For branch ℓ, the anchor is the largest apparent-power magnitude over both branch ends, Q0 and Q95, and the preselected baseline and reference operating points:

$$S_\ell^{\mathrm{anchor}} = \max_{\substack{u\in\{\mathrm{base,ref}\},\ q\in\{\mathrm{Q0,Q95}\}\\ e\in\{\mathrm{fr,to}\}}}\left|S_{\ell,e}^{u,q}\right|, \qquad \bar{S}_\ell^{\mathrm{syn}} = \max\left\{S_\ell^{\min},\ 1.10S_\ell^{\mathrm{anchor}}\right\}. \tag{S75}$$

The class floor $S_\ell^{\min}$ is 0.10 MVA for root-feeder and trunk branches and 0.05 MVA for lateral branches. The same rating is used in Q0 and Q95. Changed-request, ECF, and comparison-control flows do not enter the anchor. The rating construction was fixed before those calculations.

For fixed reactive-power setting $q_{\mathrm{pf}} \in \{\mathrm{Q0}, \mathrm{Q95}\}$, Q0 denotes zero participant reactive power. Q95 denotes 0.95 power factor with reactive-power absorption. The proxy expresses each branch-end flow

and bus-voltage magnitude as an affine function of the allocation:

$$\begin{aligned}\widehat{P}^{q_{\rm pf}}_{\ell,e}(\mathbf{x}) &= \widehat{P}^{0,q_{\rm pf}}_{\ell,e} + (\mathbf{c}^{P,q_{\rm pf}}_{\ell,e})^{\mathsf{T}}\mathbf{x},\\ \widehat{Q}^{q_{\rm pf}}_{\ell,e}(\mathbf{x}) &= \widehat{Q}^{0,q_{\rm pf}}_{\ell,e} + (\mathbf{c}^{Q,q_{\rm pf}}_{\ell,e})^{\mathsf{T}}\mathbf{x},\\ \widehat{v}^{q_{\rm pf}}_{b}(\mathbf{x}) &= \widehat{v}^{0,q_{\rm pf}}_{b} + (\mathbf{c}^{V,q_{\rm pf}}_{b})^{\mathsf{T}}\mathbf{x},\end{aligned} \tag{S76}$$

where $e \in \{\mathrm{fr}, \mathrm{to}\}$. Hats distinguish proxy predictions from the nonlinear AC quantities $P_{\ell,e}$, $Q_{\ell,e}$, the voltage phasor $\mathcal{V}_b$, and its magnitude $v_b = |\mathcal{V}_b|$; main Methods checks voltage magnitude. Superscript 0 denotes the affine intercept, and each $\mathbf{c}$ is the corresponding fixed AC-derived response vector. The intercept is evaluated at zero participant export with the fixed feeder load. Each response column is a finite difference of nonlinear AC solutions with a 0.001-MW participant-export perturbation, separately for Q0 and Q95. Let $\mathbf{D}_C = \mathrm{diag}((C_i)_{i\in\mathcal{N}})$ and use ζ, rather than the WP factor ρ, for the normalized allocation radius. The first cone is

$$\|\mathbf{D}_C^{-1}\mathbf{x}\|_2 \le \zeta, \qquad 0 \le \zeta \le \zeta^{q_{\rm pf}}_{\max}. \tag{S77}$$

For each branch and both ends, write $\mathfrak{m}$ for the fixed nonnegative guard margin. Its branch apparent-power component is affine, $\mathfrak{m}^{S,q_{\rm pf}}_{\ell,e}(\zeta) = \mathfrak{m}^{S,0,q_{\rm pf}}_{\ell,e} + \mathfrak{m}^{S,1,q_{\rm pf}}_{\ell,e}\zeta$, and the corresponding conic constraint is

$$\left\| \begin{bmatrix} \widehat{P}^{q_{\rm pf}}_{\ell,e}(\mathbf{x}) \\ \widehat{Q}^{q_{\rm pf}}_{\ell,e}(\mathbf{x}) \end{bmatrix} \right\|_2 \le \bar{S}^{\rm syn}_{\ell} - \epsilon_S - \mathfrak{m}^{S,q_{\rm pf}}_{\ell,e}(\zeta), \qquad e \in \{\mathrm{fr}, \mathrm{to}\}, \tag{S78}$$

with $\epsilon_S = 2 \times 10^{-6}$ MVA. The lower-voltage rows are

$$\widehat{v}^{q_{\rm pf}}_{b}(\mathbf{x}) - \mathfrak{m}^{\mathrm{lo},q_{\rm pf}}_{b}(\zeta) \ge V_{\min} + \epsilon_V, \qquad \epsilon_V = 2 \times 10^{-6}\ \text{pu}. \tag{S79}$$

When an upper-voltage guard-margin model is active, the corresponding row is $\widehat{v}^{q_{\rm pf}}_{b}(\mathbf{x}) + \mathfrak{m}^{\mathrm{up},q_{\rm pf}}_{b}(\zeta) \le V_{\max} - \epsilon_V$. Otherwise, the calculation uses $\widehat{v}^{q_{\rm pf}}_{b}(\mathbf{x}) \le V_{\max} - \mathfrak{m}^{\rm fix}_{V} - \epsilon_V$, with $\mathfrak{m}^{\rm fix}_{V} = 5 \times 10^{-3}$ pu. All margin coefficients and $\zeta^{q_{\rm pf}}_{\max}$ are held constant across the paired cases. Here $\bar{S}^{\rm syn}_{\ell} > 0$ is the synthetic scenario MVA rating for monitored branch ℓ, and $V_{\min}, V_{\max}$ are the voltage limits.

For each residual component, the margin slope is 1.25 times the largest positive AC-minus-proxy residual divided by $\max\{\|\mathbf{D}_C^{-1}\mathbf{x}\|_2, 10^{-6}\}$ over the calibration allocations. For lower voltage the residual sign is reversed. Each active margin has an intercept of 10^{-7} in its physical unit; an upper-voltage component with fitted slope at most 10^{-12} uses the fixed guard above instead. Calibration uses preliminary allocations from all five rules on the designated calibration cases. The coefficients are then held fixed for independent checks and the request and feedback comparisons. The normalized radius is restricted to $\zeta_{\max} = 3.5$ in both Q settings, as a specified allocation-domain bound rather than an estimated convergence radius. The deposited controlled-case input contains the participant bus map, connection bounds, reference requests, affine matrices and fitted margins.

Equations (S77) and (S78) are second-order cone (SOC) constraints. Each bounds the norm of an affine expression by an affine scalar. Cone membership also requires a nonnegative right-hand side. These are the conic constraints used for controlled-study allocation.

ME Stage 1 is a second-order cone program (SOCP). ME Stage 2 and FA contain convex quadratic terms. Epigraph variables for those terms give equivalent rotated SOC representations. The model introduces no branch-current variable and is not the standard branch-flow relaxation. A proxy-feasible result therefore establishes feasibility of the specified conic allocation domain. A separate nonlinear radial AC calculation uses the same injection vector and reactive-power setting. It checks voltage, active- and reactive-power balance, and apparent power at both ends of every monitored branch. The physical check uses $0.95 \le v_b \le 1.05$ pu and the same $\bar{S}^{\rm syn}_{\ell}$ at both branch ends. These AC calculations, rather

than the proxy alone, supply the physical-feasibility result reported in the main paper. The primary implementation uses polar Newton–Raphson power flow and is cross-checked by an independently implemented radial backward/forward sweep. Both nonlinear solvers use a 10^{-8} stopping tolerance and at most 100 iterations. The cross-check requires bus active- and reactive-power mismatches no larger than 10^{-5} MW and 10^{-5} MVAr, respectively. It also compares voltage magnitudes within 10^{-5} pu, angles within 10^{-3} degrees, and branch powers, slack powers and losses within 10^{-4} MW or MVAr. Physical-screen tolerances are 10^{-6} pu for voltage and 10^{-6} MVA for branch loading.

S9 Secondary Allocation-Formulation Sensitivities

Two supplementary tests vary the network allowance and the weight on allocation equality. They describe how those formulation choices affect no feedback and mismatch-only feedback; the primary ECF comparisons use the settings specified above.

Margin retention (MR) changes the network-feasible set to $\mathcal{F}^{\mathrm{MR}}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}; \kappa)$, where $0 \le \kappa < 1$ and

$$\mathcal{F}^{\mathrm{MR}}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}; 0) = \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}). \tag{S80}$$

Rule parameters change the allocation objective or its selection rule; MR instead changes the network allowance. The no-feedback baseline sets $\psi_i = C_i$, and the case without MR sets $\kappa = 0$. In the full proxy, MR increases each active margin intercept by κ times that constraint's nonnegative slack at zero export. For a branch end this slack is $[\bar{S}^{\mathrm{syn}}_{\ell} - \epsilon_S - \|(\widehat{P}^0_{\ell,e}, \widehat{Q}^0_{\ell,e})\|_2 - \mathfrak{m}^{S,0}_{\ell,e}]^+$. For lower voltage it is $[\widehat{v}^0_b - V_{\min} - \epsilon_V - \mathfrak{m}^{\mathrm{lo},0}_b]^+$; for an active upper-voltage margin it is $[V_{\max} - \epsilon_V - \widehat{v}^0_b - \mathfrak{m}^{\mathrm{up},0}_b]^+$. Margin slopes, the radius bound and inactive upper-voltage rows are unchanged. Thus MR reserves a fraction of origin slack, not a fraction of the full branch rating.

S9.1 Retained network margin

We evaluate $\kappa \in \{0, 0.025, 0.05, 0.075, 0.10\}$ with no feedback and mismatch-only feedback. The feeder, scenario ratings, reference requests, reactive-power setting and AC formulation remain fixed. Across this range, the largest absolute change in case-level allocation movement M_x relative to $\kappa = 0$ is 0.151 W, and the largest change in branch-loading ratio is 3.01×10^{-8}. Every matched reactive-power pair passes the specified AC checks. MR has negligible effect over this tested range.

S9.2 Fairness-aware objective weight

The FA analysis varies

$$\alpha_{\mathrm{FA}} \in \{0.05, 0.10, 0.20, 0.40, 0.80\} \text{ MW}.$$

The denominator offset remains $\epsilon_{\mathrm{FA}} = 10^{-6}$ MW. At each weight, mismatch-only feedback is compared with no feedback under the same allocation formulation. Across temporal profiles, mismatch-only feedback reduces mean M_x by 18.43–34.68% and raises the capacity-normalized Jain index by 0.139–0.648 percentage points. Every matched reactive-power pair passes the specified AC checks. The primary ECF comparisons retain their FA weight.

Part III. Public-data network simulations

S10 Public Data and Scenario Construction

The public-data analyses test whether the delivery opportunity identified in a completed interval can guide useful restrictions in later intervals. Public PV and load histories provide varying output, and the

network model determines how requests become awards. Section S11 first compares delivered energy, then tests whether similarly frequent replay-free updates reproduce the benefit. Section S12 examines response strength, memory and record error. Sections S13–S14 vary electrical placement, PV capacity, network structure and participants' requests, and report how the aggregate gain is distributed.

S10.1 Public network and household records

The public-data simulation combines the Australian MV/LV OpenDSS model reported by Geth *et al.*[9] with the Ausgrid Solar Home Electricity Data[10]. The network contains one radial 11-kV feeder, 33 associated low-voltage sections, 33 transformers, and 1,255 service connections; 30 LV sections carry mapped service loads. Its topology and several equipment properties derive from geographic information system (GIS) records, with documented engineering completions for grounding, impedance mapping, and connection phases. Transformer nameplates are available, whereas usable native line ampacities are not.

The collection also provides executable models for individual LV sections. We use 13 load-bearing sections to examine spatial variation within the parent network under a common chronology, mapping rule, and scenario.

The Ausgrid source reports half-hour gross generation (GG), general consumption (GC), and controlled load (CL) for 300 gross-metered solar homes from 1 July 2010 to 30 June 2013. The source unit is kWh per half-hour. Division by 0.5 h gives average kW, and division by 1000 gives the MW quantities used below.

The household and network sources have no common customer identifier. The simulation therefore uses the deterministic profile-to-service mapping defined in Section S10.4. Section S10.3 defines source-record and calendar eligibility.

Supplementary Table 10: Public sources and fixed scenario construction for the Australian simulation

Element	Representation in the simulation
Network	One published unbalanced MV/LV model with GIS-derived topology and documented engineering completions
Chronology	Public half-hour gross PV and household load records
Mapping	Deterministic fixed order from profiles to service points, with cyclic reuse only after all profiles have been used
Participants	418 scenario export participants among 1,255 load connections; the remaining connections are non-participating load
PV scale	Scenario mean of 9.9 kWp across the 300 source profiles and 9.784 kWp after fixed reuse across 418 participant connections
Request	Previous-day same-half-hour proxy plus the positive part of a 90th-percentile day-to-day change estimated from 2010–2011, bounded by the scenario connection limit
Post-event record	Net-available-export proxy constructed offline from scaled gross-PV and household-consumption records
Network calculation	Unbalanced nonlinear AC with customer-voltage and native transformer-nameplate limits

The SI, calculated results, code, source identifiers, preprocessing steps, retained-date rules, and deterministic profile-to-service mapping are deposited in the public project repository, `https://github.com/MarxHuang/ecf-export-access`. Original measurements and network files are obtained from the cited distributors under their applicable terms. The repository instructions distinguish checks of saved result tables from full simulation reruns and identify the external inputs and auxiliary trajectory files required for each analysis.

S10.2 Information used by the completed-state replay

ECF relates four time-aligned quantities. Before an interval, a participant or aggregator submits an export request. The network allocation determines authorized access, which may be issued as an operating envelope. After the interval, metering records the delivered export. A post-event available-export record estimates how much export was available during that completed interval under the same-state assumption. In the Australian simulation, the request is a constructed forecast, authorized access and delivery are model outcomes, and the post-event record is constructed from gross-PV and household-consumption profiles.

In an operational implementation, these quantities must share a connection or portfolio identifier and timestamp. The post-event record may come from a controller or inverter with authenticated telemetry, or an independently validated estimate of available export in the completed interval. A participant's unverified forecast alone does not establish that quantity. Curtailment status and feeder location are also needed when they determine whether the same-state assumption applies. The allocation process need not receive raw household load and PV traces if a validated available-export value is supplied.

The public sources provide household profiles and a network model rather than a linked operational panel. The simulation therefore constructs the quantities in Table 11 and tests perturbations to the post-event record. Project EDGE illustrates operational interfaces for operating envelopes, real and reactive power, and aggregator exchange[11]; a linked available-export record remains an additional requirement for an operational ECF study.

Supplementary Table 11: Information used by ECF and its representation in the Australian simulation

Quantity	Operational role	Representation in the simulation
Request or contribution plan	Participant or aggregator input before allocation	Forecast request constructed from the public profiles
Issued access or dynamic operating envelope (DOE)	Distribution-network operator authorization after network assessment	Output of the allocation model
Completed-interval delivery y_i^c	Connection-point export under control c	Control-dependent model output $\min\{x_i^c, a_i^{\mathrm{prof}}\}$; it is not source-metered export
Available-export record and constraint status	Input to the completed-state replay	Primary record $a_i^{\mathrm{tel}} = a_i^{\mathrm{prof}}$; sensitivity cases alter only a_i^{tel}
Connection-to-feeder identity	Network location of each participant	Deterministic cross-source profile mapping

The completed-state comparison requires the available-export record to have a known origin and time alignment. Section S12.1 tests how the result changes when the profile-derived record is scaled, delayed, or partly unavailable.

S10.3 Chronological roles and record selection

The three source years have separate chronological roles. Data from 2010–2011 estimate the request-construction quantile. Data from 2011–2012 select the feeder load multiplier from a fixed candidate set and qualify the allocation-stage upper-voltage ceiling. Only the 2012–2013 trajectories contribute to the reported paired outcomes. We refer to these periods as request construction, feeder setting, and assessment, respectively.

A blank quality field identifies an actual record in the Ausgrid source. Rows labelled "NA" are estimated or substituted and are excluded. An eligible customer-day contains actual GC and GG records. If that customer-year includes a CL channel, its CL record must also be actual; a missing CL value is not treated as zero.

Eligibility is assessed over the unique source profiles mapped to all modeled load connections, including nonparticipants; cyclic reuse does not create another completeness requirement. Every retained

day and its previous calendar day must contain complete, chronology-valid records for all such profiles. For the first day of a chronological split, the previous-day check uses the final calendar day in the preceding split. Six daylight-saving transition dates are excluded because the source retains 48 local-clock entries without a unique UTC chronology. The primary mapping leaves 157 feeder-setting days and 80 assessment days. The assessment days span 2 January to 27 June 2013: 21 days in January, 11 in April, 24 in May, and 24 in June. No day is eligible in July–December 2012 or February–March 2013. All reported energy totals therefore refer to the retained half-hours, not to a complete year.

A separate sensitivity retains 255 customers with complete records in all three years under the same 0.30 load multiplier. This sensitivity changes customer composition and the resulting calendar coverage while keeping the load multiplier fixed. The remapped profiles can still change the realized feeder load.

Common paired-day inference. The statistical unit is one paired network day. Paired-day percentile intervals use 10,000 resamples. A second interval allows short-range dependence between nearby eligible dates. Let $\Delta\mathcal{E}_d$ be the paired energy difference on date $d \in \mathcal{D}$, with sample mean $\overline{\Delta\mathcal{E}}$. The integer calendar-day separation is $\Lambda_{d,d'} = |d - d'|_{\text{days}}$. For bootstrap replicate s_{boot}, draw a Gaussian multiplier vector $\boldsymbol{\varpi}^{(s_{\text{boot}})} = (\varpi_d^{(s_{\text{boot}})})_{d\in\mathcal{D}}$ with covariance

$$\operatorname{Cov}(\varpi_d, \varpi_{d'}) = \max\left\{1 - \frac{\Lambda_{d,d'}}{8}, 0\right\}. \tag{S81}$$

The dependent wild-bootstrap replicate is

$$\overline{\Delta\mathcal{E}}^{(s_{\text{boot}})} = \overline{\Delta\mathcal{E}} + \frac{1}{|\mathcal{D}|}\sum_{d\in\mathcal{D}} \varpi_d^{(s_{\text{boot}})}(\Delta\mathcal{E}_d - \overline{\Delta\mathcal{E}}). \tag{S82}$$

We use 10,000 antithetic replicates and the 0.025 and 0.975 quantiles. The Bartlett covariance decreases linearly with the actual calendar-day gap: it is positive at separations of 1–7 days and zero at 8 days or more. Every eligible day has equal weight. The resulting interval describes the mean over the observed eligible dates while allowing short-range dependence.

The primary ECF-minus-no-feedback gain is 17.4 kWh per assessed day. Repeating the date-dependent calculation with correlation supports of 4, 8, 14 and 21 calendar days gives respective 95% intervals of 5.4–29.4, 4.9–29.9, 5.3–29.5 and 5.2–29.6 kWh per day. This bandwidth comparison uses a separate Gaussian draw with seed 2448. The primary interval in the main text, 5.1–29.7 kWh, uses the stored primary calculation with seed 20260813; both use the same 8-day covariance support. Their small difference is Monte Carlo variation, not rounding or a different effect estimate. The longest uninterrupted 35-day block contributes 1.064 MWh of the 1.392-MWh total. On the other 45 days the difference is 0.329 MWh, or 7.3 kWh per day, with an 8-day-support interval of −4.9 to 19.5 kWh per day. This removal is a sensitivity check; the primary estimate retains all 80 eligible days. The corresponding equal-total contrasts differ by less than 0.01 kWh per day. The calculation script and daily input table accompany the deposited results.

S10.4 Scenario construction and network allocation

Let $\iota(i)$ map service connection i to one source profile. The mapping is deterministic and covers all 1,255 service connections. The forward-uptake scenario assigns export participation to 418 connections, or approximately one third of the total. Each historical gross-generation profile is divided by its documented source capacity and rescaled to its scenario PV capacity. For this scenario, the export connection bound C_i is set numerically equal to that PV capacity; it bounds admitted requests and authorized access. Household active demand is GC+CL, uses 0.95 lagging power factor, and is multiplied by the selected global load scale. PV export operates at unity power factor. For allocation, this retrospective simulation

treats contemporaneous half-hour household demand as fixed exogenous data; load-forecast error is not modeled.

All 1,255 service connections receive household profiles, including those without export participation. Connections and the 300 customer identifiers are placed in separate fixed hash orders, paired in order, and the customer list is cycled only after all profiles have been used. The first 418 connections in the fixed adoption ranking receive PV and export participation; nonparticipants retain mapped household demand but have no modeled PV. The deposited mapping table identifies every connection, profile and rank.

For the AC equations, let L_i and G_i be scaled household demand and gross PV power in MW. For nonparticipating connections, set $G_i = x_i = y_i = 0$ while retaining household demand and reactive load. Positive active power denotes consumption. The no-PV load screen uses $P_i = L_i$ at every connection. The authorization screen uses $P_i = -x_i$ at participants and $P_i = L_i$ elsewhere. The delivery screen uses

$$P_i = [L_i - G_i]^+ - \min\{x_i, [G_i - L_i]^+\}.$$

Household reactive demand is $Q_i = L_i \tan(\arccos 0.95)$ in each screen; PV has unity power factor. Thus local self-consumption is accounted for before net export, without subtracting household demand a second time. The primary replay-delivery screen substitutes the replay award for x_i. The record-driven screen in Section S12.3 instead combines completed residual import with the estimated replay export.

Here d indexes calendar days and $\tau \in \{1, \ldots, 48\}$ indexes retained half-hours within a day.

The gross-PV scale is $\lambda_{\mathrm{PV}} = 5.8813$; the repository input retains full numerical precision. The selected load multiplier is $\lambda_L = 0.30$ for all public-data cases. For participant i, the profile-derived available-export proxy on calendar day d and half-hour τ is

$$a_i^{\mathrm{prof}}(d,\tau) = \left[\lambda_{\mathrm{PV}}\, \mathrm{GG}_{\iota(i)}(d,\tau) - \lambda_L\big(\mathrm{GC}_{\iota(i)}(d,\tau) + \mathrm{CL}_{\iota(i)}(d,\tau)\big)\right]^+. \tag{S83}$$

For control c, modeled connection-point delivery is

$$y_i^c(d,\tau) = \min\{x_i^c(d,\tau), a_i^{\mathrm{prof}}(d,\tau)\}. \tag{S84}$$

The primary replay record is $a_i^{\mathrm{tel}}(d,\tau) = a_i^{\mathrm{prof}}(d,\tau)$. Record sensitivities change only a_i^{tel}. The proxy $a_i^{\mathrm{prof}}(d,\tau)$ can exceed C_i. The connection bound limits the capacity-limited and admitted requests and authorized access, not underlying availability: $r_i^{\mathrm{cap}}, r_i^{\mathrm{adm}}, x_i \le C_i$ and $y_i \le x_i$. The proxy is constructed from the public profiles; it is not a direct measurement of latent a_i. Let $\mathcal{D}_{\mathrm{req},\tau}$ be the eligible residual-estimation day pairs with valid current- and previous-day observations at half-hour τ. The request forecast adds a nonnegative residual-quantile allowance to the previous-day value:

$$r_i(d,\tau) = a_i^{\mathrm{prof}}(d-1,\tau) + \left[Q_{0.90,\, d' \in \mathcal{D}_{\mathrm{req},\tau}}\left(a_i^{\mathrm{prof}}(d',\tau) - a_i^{\mathrm{prof}}(d'-1,\tau)\right)\right]^+, \tag{S85}$$

$$r_i^{\mathrm{cap}}(d,\tau) = \min\{r_i(d,\tau), C_i\}. \tag{S86}$$

The signed residual quantile is evaluated first, and only its positive part is added to the previous-day value. For m finite residuals sorted as $e_{(1)} \le \cdots \le e_{(m)}$, the empirical quantile uses linear interpolation at $h = (m-1)0.90$: with $k = \lfloor h \rfloor$ and $\lambda = h - k$, it is $(1-\lambda)e_{(k+1)} + \lambda e_{(k+2)}$, with the endpoint convention when $k = m-1$. This is the linear empirical-quantile convention used in the calculations. The first relation defines the previous-day-plus-residual raw request. The second applies the fixed connection bound. The admitted request then follows Eq. (S1) through the later request ceiling. At the start of each daily trajectory, $\psi_i(1) = C_i$.

For each half-hour, the Australian allocation applies a common factor $\rho \in [0,1]$ to the admitted requests. Zero-anchored bisection yields the largest feasible lower-bracket factor encountered. This is a feasible-factor search; global maximality would additionally require monotonic feasibility or a search of

the full factor interval. The search stops when the factor bracket is at most 10^{-5} or after 24 iterations. At every candidate, the load-only and authorized-export endpoints must converge, remain above 0.90 pu, and satisfy native transformer nameplates. The authorized endpoint must also remain below the 1.07-pu allocation-stage upper-voltage ceiling. The profile-derived delivery state in (S84) is separately screened against the 0.90–1.10-pu physical voltage range and the same transformer limits. Customer voltage is the voltage across each modeled service terminal, using phase-to-neutral values for wye connections and phase-to-phase values for delta connections. Every represented phase is checked. The OpenDSS nonlinear solution tolerance is 10^{-10}.

An independent factor-grid check covered all 19,200 allocation calls in the primary 80-day study: the four control trajectories and ECF's completed-state replays. Of these, 11,349 accepted the full request at $\rho = 1$. Each remaining call was checked at 65 equally spaced factors over $[0, 1]$, augmented by 17 points from the returned factor to 1 and points 10^{-5}, 4×10^{-5}, and 16×10^{-5} above it, capped at 1. The returned factor was also included. All 7,851 constrained calls retained a feasible returned factor. No feasible sampled point lay above it, and feasibility did not reappear after a failed sampled point. These checks used the same loads, requests, and network limits, after completing each original daily trajectory. They support the primary allocations at the sampled resolution, not a proof of global optimality or a full-grid check of the other scenarios.

The scenario uses a load multiplier of 0.30 and a 1.07-pu authorization voltage ceiling, leaving 0.03 pu below the 1.10-pu physical upper limit. No-feedback operation on 157 setting dates was used to find conditions that retain useful export and intervals with constrained requests while satisfying the physical limits in each of the 13 LV sections. All 30 reported public-data cases retain these load and voltage settings; modeled-delivery states are screened separately. The complete-customer cohort changes composition and calendar coverage under the same load multiplier.

Initial exploratory runs informed the operating setting. The reported ECF and equal-total outcomes did not change its parameters; both use the same setting throughout the 80-day comparison. The fixed-update, cross-day-memory, profile-placement and PV-capacity tests were developed subsequently and are interpreted as sensitivity analyses. The repository records the scenario versions and construction sequence.

The four controls are no feedback, previous-day persistence, ECF with $\beta_{\mathrm{ECF}} = 0.5$, and the equal-total control. Previous-day persistence uses the preceding day's same-half-hour available-export proxy with no residual-quantile increment and no replay. It tests a more conservative request forecast than the other public controls. The equal-total control removes the same total request as ECF in each half-hour. Its paired difference from ECF therefore isolates where that reduction is applied.

The Australian simulation uses the unused-award and replay tolerances

$$\epsilon_x^{\mathrm{AU}} = \tau_{\mathrm{rpl}}^{\mathrm{AU}} = 2 \times 10^{-6} \text{ MW}. \tag{S87}$$

The same value defines an unused award and each strict replay comparison. Table 6 compares these thresholds with the controlled-study values and distinguishes them from record-error bounds.

For the post-event-record sensitivities, every inferred record is bounded below by current modeled delivery. With $\lambda_{\mathrm{rec}} \in \{0.9, 1.1\}$,

$$a_i^{\mathrm{tel},\lambda_{\mathrm{rec}}}(d,\tau) = \max\{y_i(d,\tau), \lambda_{\mathrm{rec}} a_i^{\mathrm{tel}}(d,\tau)\}, \tag{S88}$$

$$a_i^{\mathrm{tel,lag}}(d,\tau) = \begin{cases} \max\{y_i(d,\tau), a_i^{\mathrm{tel}}(d,\tau-1)\}, & \tau > 1, \\ y_i(d,1), & \tau = 1. \end{cases} \tag{S89}$$

The deterministic 10% missing-record pattern replaces each omitted record with $y_i(d,\tau)$ and leaves other records unchanged. Since $y_i(d,\tau) \le a_i^{\mathrm{prof}}(d,\tau)$, this substitution gives a lower bound on replay delivery, not the missing availability itself. Remaining records can establish a replay delivery gain; otherwise, ceilings rise toward capacity. Transformed records remain at least as large as modeled delivery.

These perturbations alter only the record used to evaluate replay recovery. Control-dependent delivery remains generated from the unperturbed a_i^{prof}. For each sensitivity, its equal-total control is rebuilt from that sensitivity's ECF admitted-request total. The public physical assessment continues to check the load-only and authorized endpoints plus the delivery state formed from replayed access and unperturbed a_i^{prof}; a transformed a_i^{tel} is not introduced as a separate AC injection. The 10% unavailability pattern is selected deterministically among participant records using the fixed seed recorded in the archived sensitivity configuration.

Computational workload. Across the 3,840 primary half-hours, ECF uses 80,941 power-flow calls, including its allocation, delivery checks, and completed-state replays. No feedback and the equal-total control each use 43,628 calls; previous-day persistence uses 37,201. ECF therefore averages 21.1 calls per half-hour, compared with 11.4 without feedback. The implementation attempts a replay in every half-hour, although only 483 comparisons lead to tightening. The frequency of tightening does not measure the frequency of replay computation. These counts describe the simulation workload, excluding model initialization and the independent numerical-search checks; they do not measure execution time or deployment cost.

S11 Australian Simulation: Energy and Control Comparisons

S11.1 Energy accounting and the primary delivery result

ECF delivers 1.392 MWh more modeled export than each of no feedback and the equal-total control. It both authorizes more export and leaves less of that authorization unused. The energy balance below decomposes the delivery difference into changes in authorization and its unused portion, distinguishes unused permission from available electricity not delivered, and traces the gain to source-terminal exchange. Each control uses the same calendar and available-export profiles.

Across Sections S11–S14, paired network days are the statistical comparison unit. Random schedules and request-choice seeds are averaged within each date where stated. Reused dates, profiles and control trajectories are not counted as additional independent observations; spatial scenarios are reported separately rather than pooled across feeders. Section S14.3 collects physical-check coverage and the shared interpretation of these tests.

For control c and an eligible set $\mathcal{D}$ of network days, define

$$
\begin{aligned}
E_Y^c &= 0.5 \sum_{d\in\mathcal{D}} \sum_{\tau=1}^{48} \sum_i y_i^c(d,\tau), \\
E_X^c &= 0.5 \sum_{d\in\mathcal{D}} \sum_{\tau=1}^{48} \sum_i x_i^c(d,\tau), \qquad E_H^c = E_X^c - E_Y^c, \\
E_{\text{avail}} &= 0.5 \sum_{d\in\mathcal{D}} \sum_{\tau=1}^{48} \sum_i a_i^{\text{prof}}(d,\tau), \qquad E_{\text{avail}-Y}^c = E_{\text{avail}} - E_Y^c,
\end{aligned}
\tag{S90}
$$

where power is in MW and energy is in MWh. The profile-derived available-export proxy is common to all controls, so E_{avail} has no control superscript. The quantity E_H^c is unused authorization. By contrast, $E_{\text{avail}-Y}^c$ is profile-derived available export that is not delivered; it is nonnegative because $y_i^c \le a_i^{\text{prof}}$. These quantities are summed at participating connections after on-site self-consumption, but before network losses and consumption elsewhere in the feeder. To evaluate the network endpoint, let $P_{\text{src}}^c(d,\tau)$ be the OpenDSS source-terminal active power. Let $P_{\text{loss}}^c(d,\tau)$ be the recorded circuit loss. Both are converted to MW. Under the OpenDSS sign convention for this quantity, negative P_{src} denotes

power supplied to the feeder and positive P_{src} denotes export toward the upstream system[12]. Define

$$E^{c}_{\mathrm{up}} = 0.5 \sum_{d \in \mathcal{D}} \sum_{\tau=1}^{48} P^{c}_{\mathrm{src}}(d,\tau),$$
$$E^{c}_{\mathrm{loss}} = 0.5 \sum_{d \in \mathcal{D}} \sum_{\tau=1}^{48} P^{c}_{\mathrm{loss}}(d,\tau), \tag{S91}$$

and, for a target control c and comparison control c_0,

$$\Delta E_{\mathrm{adj}}(c, c_0) = \left(E^{c}_{Y} - E^{c_0}_{Y}\right) - \left(E^{c}_{\mathrm{up}} - E^{c_0}_{\mathrm{up}}\right). \tag{S92}$$

A negative E^{c}_{up} denotes net import over the eligible panel. The paired adjustment contains changes in circuit loss and voltage-dependent passive-element exchange, so E_{loss} is reported separately. Table 12 gives the connection-point energy totals.

Paired-day intervals use the common procedure in Section S10.3. The energy definitions above apply to every control; the following tables report their totals and paired contrasts.

Supplementary Table 12: Australian simulation over 80 assessment days

Control	E^{c}_{X}	E^{c}_{Y}	E^{c}_{H}	$E^{c}_{\mathrm{avail}-Y}$
	MWh over assessed half-hours			
No feedback	328.490	237.963	90.526	655.765
Previous-day persistence	212.296	189.030	23.267	704.699
ECF	329.696	239.356	90.341	654.373
Equal-total control	328.483	237.964	90.519	655.765

Relative to no feedback, ECF increases authorization by 1.206 MWh and reduces unused authorization by 0.186 MWh. The identity $\Delta E_Y = \Delta E_X - \Delta E_H$ gives a delivery increase of 1.392 MWh. Relative to the equal-total control, the corresponding changes are +1.213 and −0.179 MWh, giving a delivery increase of 1.392 MWh to the displayed precision. In this case, increased authorization and reduced unused awards both contribute to the accounting identity in (S51). This decomposition is an energy balance, not an identification of two independent causal effects. Equal-total admitted requests can yield different award totals through the network-dependent proportional factors in (S56).

Supplementary Table 13: Network-endpoint and access-use quantities over the same 80 assessment days. Positive E_{up} denotes net export toward the upstream source; all four totals are negative and therefore denote net import.

Control	E^{c}_{up} MWh	E^{c}_{loss} MWh	E^{c}_{Y}/E^{c}_{X} %	Constrained intervals %
No feedback	−260.959	2.874	72.44	44.43
Previous-day persistence	−309.584	2.662	89.04	34.95
ECF	−259.582	2.885	72.60	44.40
Equal-total control	−260.959	2.874	72.44	44.43

Here E^{c}_{Y}/E^{c}_{X} is the fraction of authorized access that is delivered. The final column is the fraction of half-hours in which the common request scale is below one by more than the stated numerical tolerance. Previous-day persistence has the highest utilization ratio, but it also has the lowest delivered export.

Against no feedback, ECF's mean daily delivery increase is 0.0174 MWh. Its ordinary paired 95% interval is [0.0061,0.0307] MWh per network day, and its dependent wild-bootstrap interval is

Supplementary Table 14: Paired network-endpoint differences for ECF over the eligible panel. Every numeric contrast is ECF minus the comparator named in its row. The final column is the dependent wild-bootstrap 95% interval for the mean daily change in E_{up}.

Comparator	Delivered	Source exchange	Circuit loss	Adjustment	Daily source-exchange interval
	MWh over assessed half-hours				MWh per network day
No feedback	1.392	1.377	0.010	0.015	[0.0046, 0.0298]
Equal-total control	1.392	1.377	0.010	0.015	[0.0049, 0.0295]

[0.0051,0.0297]. Against the equal-total control, the mean is also 0.0174 MWh per day; the intervals are [0.0060,0.0306] and [0.0049,0.0299], respectively. Each comparison has 44 positive days, 32 negative days, and four zero days. These intervals characterize the mean paired change.

The corresponding source-terminal increases average 0.0172 MWh per network day. Their dependent wild-bootstrap intervals are [0.0046,0.0298] against no feedback and [0.0049,0.0295] against the equal-total control. Source-terminal gains retain 98.9% of both connection-point gains. Circuit losses increase by about 0.0102 MWh in each comparison. The remaining delivery–source differences are about 0.0048 MWh and include changes in voltage-dependent passive-element exchange.

All four replay conditions hold in 483 of 3,840 half-hours (12.578%). Within these intervals, completed-state delivery recovery has a median of 3.276 kW and an interquartile range of 0.136–14.618 kW. Modeled-delivery voltages range from 0.9363 to 1.0736 pu, and maximum transformer loading is 0.2736 pu. The replay's physical checks separately qualify the states used to trigger updating.

S11.2 Replay qualification with a fixed update rule

ECF and a replay-free control use the same ceiling-update formula and gain. The control omits the completed-state replay and uses the target in (S36) with the replay indicator replaced by one: each member of its own current unused-award group targets completed delivery, and every other participant targets its fixed capacity. Both controls apply (S37) with gain 0.5. They independently recompute allocation, delivery, and the next ceiling from their own evolving state. The control without qualification needs no replay allocation. It is distinct from mismatch-only feedback, which smooths a delivery-to-award ratio.

The comparison uses all 80 primary assessment days, 418 export participants, 48 half-hours per day, and the same daily capacity reset, raw requests, available export, mapping, network, allocation search, and physical limits. The ECF branch reproduces the primary request, award, delivery, and ceiling trajectories. Only the use of replay qualification differs between controls.

ECF gains 6.423 MWh of modeled delivery, or 2.76%, relative to updating after every unused award. The daily difference is positive on 75 days and negative on five. Its mean is 80.3 kWh per network day; the paired percentile 95% interval is 68.0–92.3 kWh and the date-dependent wild interval is 69.3–91.2 kWh. Both use the paired-day procedures above with 10,000 replicates. Across participants, 417 have higher total delivery under ECF and one has lower delivery, by 1.17 kWh. The median participant difference is 12.57 kWh, with an interquartile range of 7.70–19.86 kWh.

The control without qualification reduces unused authorization more, but its larger reduction in authorization also removes useful delivery. The energy identity gives $24.537 - 18.114 = 6.423$ MWh for ECF's advantage. Net source exchange increases by 6.374 MWh, reflecting reduced upstream import.

Testing the delivery opportunity before applying the same update therefore preserves useful authorization in this comparison. It also reduces the number of updates. The next subsection asks whether randomly selecting a similar number of updates can recover the same benefit. Section S12.2 then examines ceiling memory across days.

Supplementary Table 15: Replay qualification with the ECF update held fixed. Energy totals cover 80 network days. Differences are ECF minus ECF without replay qualification and are calculated before rounding.

Quantity	ECF	Without qualification	Difference
Delivered export (MWh)	239.356	232.933	6.423
Authorized export (MWh)	329.696	305.159	24.537
Unused authorization (MWh)	90.341	72.226	18.114
Source-terminal exchange (MWh)	−259.582	−265.956	6.374
Circuit losses (MWh)	2.885	2.850	0.035
Delivery-target intervals	483	3,086	−2,603
Participant ceiling reductions	73,983	359,263	−285,280
Power-flow calls	80,941	42,251	38,690
Physically passing half-hours	3,840	3,840	0

S11.3 Update selection at a comparable frequency

The preceding comparison changes both the information used to select an interval and the frequency of ceiling reductions. The following control tests interval selection more closely by choosing intervals without replay at approximately ECF's update frequency. It uses the same 80-day simulation and a probability calibrated from the primary ECF result. This is a post-hoc comparison of update selection, rather than an independently chosen operating policy.

The primary ECF trajectory had an unused-award group in 3,091 half-hours and qualified 483 of them. We therefore set $p = 483/3091$ before running the random control. For seed s, date d and half-hour t, a fixed SHA-256 hash of (s, d, t) gives $u_s(d, t) \in [0, 1)$. The replay-free decision is

$$g_s^{\mathrm{rnd}}(d, t) = \mathbf{1}\{\mathcal{N}_{\mathrm{ua},s}(d, t) \neq \varnothing\}\mathbf{1}\{u_s(d, t) < p\}. \tag{S93}$$

The draw does not use the completed-state replay or the post-event available-export record. Eligibility still uses the control's own completed award and delivery. When selected, the unused-award group's target is its completed delivery; otherwise all targets are connection capacities. The ceiling formula and gain are exactly those of ECF. We use 12 fixed seeds, 2401–2412. All schedules contribute to the reported mean. Each has its own evolving awards, delivery and ceilings. Raw requests, exogenous profiles, network, allocation solver, physical limits and daily ceiling initialization are shared with ECF.

ECF delivers 0.917–1.596 MWh more than each of the 12 fixed schedules. Against their within-date average, it gains 1.343 MWh, or 16.8 kWh per network day. The date-dependent wild-bootstrap 95% interval for the daily mean is 7.8–25.8 kWh; the comparison is positive on 66 dates and negative on 14. The longest continuous 35-day block contributes 0.853 MWh of this contrast, and the other 45 dates contribute 0.490 MWh. The replay-free controls select delivery targets in 463–519 half-hours, compared with 483 for ECF. Ceiling recovery still operates in other intervals.

The replay-free schedules leave 90.345 MWh of authorization unused on average, almost the same as ECF's 90.341 MWh. ECF instead authorizes 1.339 MWh more and delivers 1.343 MWh more. This accounting comparison points to useful access preserved by the selected updates, rather than a smaller idle-award total. Matching update frequency approximately does not fix the magnitude or duration of restrictions: these follow from each control's evolving awards, delivery and ceilings. The source data retain all 12 schedules and paired daily contrasts.

Supplementary Table 16: Replay-free updates selected at approximately ECF's frequency over 80 network days. Energy totals are in MWh. The final column is ECF minus the specified randomized schedule; the mean row averages all 12 schedules. All controls pass the stated allocation and physical screens.

Control	Delivery-target intervals	Authorized	Unused	Delivered	ECF minus control
ECF	483	329.696	90.341	239.356	–
Random seed 2401	484	328.456	90.513	237.943	1.413
Random seed 2402	495	328.386	90.116	238.270	1.086
Random seed 2403	489	328.099	90.324	237.775	1.580
Random seed 2404	504	328.049	90.263	237.785	1.570
Random seed 2405	475	328.385	90.223	238.161	1.194
Random seed 2406	481	328.316	90.452	237.864	1.492
Random seed 2407	473	328.335	90.277	238.058	1.298
Random seed 2408	498	328.105	90.316	237.789	1.566
Random seed 2409	463	328.840	90.401	238.439	0.917
Random seed 2410	468	328.545	90.400	238.144	1.211
Random seed 2411	519	328.699	90.539	238.160	1.196
Random seed 2412	506	328.071	90.312	237.759	1.596
Random mean	487.9	328.357	90.345	238.012	1.343

S12 Temporal Memory and Record Sensitivity

S12.1 Gain and input sensitivities around the common scenario

The tested record perturbations preserve positive mean delivery differences, whereas unit gain changes the sign of the aggregate result. Identifying a past delivery opportunity is therefore only one part of the feedback problem; the ceiling response must also leave room for subsequent output. Table 17 reports these comparisons and the complete-customer cohort, with uncertainty intervals for each mean.

The multiplicative, lagged and missing-record tests perturb the recovery calculation while retaining unperturbed profile-derived availability for physical assessment. Section S12.3 separately evaluates a record-based decision screen, followed by a reference check using unperturbed availability.

The gain and record cases use the same 80-day panel, feeder settings, request forecast, network, and controls as the primary simulation. The complete-cohort case retains 255 source customers with records in all three source years. It uses the same mapping rule and 0.30 load multiplier. It changes customer composition and the retained dates while keeping the loading assumption fixed.

Supplementary Table 17: Australian simulation sensitivity outcomes. Values are ECF-minus-comparator paired network-day mean delivered-export differences in MWh, with dependent wild-bootstrap 95% intervals in brackets. The final column indicates whether the specified allocation and modeled-delivery checks pass throughout all four control trajectories.

Setting	Days	β_{ECF}	No feedback	Equal-total control	All checks pass
Primary unperturbed proxy	80	0.50	0.0174 [0.0051, 0.0297]	0.0174 [0.0049, 0.0299]	Yes
Lower gain	80	0.25	0.0168 [0.0043, 0.0293]	0.0168 [0.0042, 0.0294]	Yes
Unit gain	80	1.00	−0.0380 [−0.0456, −0.0305]	−0.0067 [−0.0148, 0.0013]	Yes
Record ×0.9	80	0.50	0.0164 [0.0040, 0.0287]	0.0163 [0.0037, 0.0290]	Yes
Record ×1.1	80	0.50	0.0176 [0.0058, 0.0294]	0.0176 [0.0057, 0.0296]	Yes
One-interval lag	80	0.50	0.0180 [0.0067, 0.0294]	0.0180 [0.0067, 0.0293]	Yes
10% records unavailable	80	0.50	0.0171 [0.0047, 0.0295]	0.0171 [0.0046, 0.0296]	Yes
Complete cohort	361	0.50	0.0304 [0.0214, 0.0393]	0.0304 [0.0217, 0.0390]	Yes

All settings use load multiplier 0.30 and authorization voltage ceiling 1.07 pu. The complete cohort changes customer composition and calendar coverage; the other rows change the indicated gain or record setting.

The completed-interval comparisons also differ in their margin to record error. For each of the 483 qualifying primary intervals, we increase a common absolute error radius, equal for all participants, in (S34). The original allocation, replayed allocation, unused-award group, and completed delivery are held fixed. The limiting radius is the supremum for which both the outside-delivery and signed total-delivery lower bounds exceed the original 2 W recovery threshold. The median limiting radius is 1.88 kW per participant; the fifth percentile is 90.6 W and the minimum is 4.8 W. Below each interval's limiting radius, both delivery conditions remain satisfied for all available-export values consistent with the componentwise error bound and completed delivery. At the boundary, at least one strict recovery condition ceases to hold.

This calculation distinguishes tolerance of the delivery comparison from the 2 W numerical threshold. The wide spread shows why a positive mean gain under the tested perturbations does not establish the reliability of every individual comparison. It assumes exact completed delivery and allocation-invariant availability, as in Section S3. It evaluates fixed recorded pairs rather than a new feedback trajectory, and does not establish an instrument specification or AC feasibility throughout the availability-error range.

Each one-at-a-time record perturbation has a positive dependent wild-bootstrap interval against both comparators. Unit gain reduces total delivery by 3.043 MWh relative to no feedback and 0.537 MWh relative to the equal-total control. Its dependent interval lies below zero in the first comparison and includes zero in the second. Thus the negative total against the equal-total control does not establish a statistically resolved negative mean.

The authorization–delivery identity explains the energy balance of this reversal. Relative to no feedback, unit gain reduces authorization by 14.549 MWh and unused authorization by 11.506 MWh, leaving a delivery loss of 3.043 MWh. At gain 0.25, authorization instead increases by 1.482 MWh and unused authorization by 0.137 MWh, giving a delivery gain of 1.345 MWh. All differences use the same 80 dates and are calculated before rounding. Thus a reduction in unused authorization is neither sufficient nor necessary for an aggregate delivery gain in these comparisons.

Individual ceiling trajectories locate the delivery differences in time. They are reconstructed from the recorded allocation factors and replay indicators, the original requests and available export, and the update in (S37). Their interval totals agree with the original records for all three gains. Relative to gain 0.5, unit gain loses a net 5.471 MWh over 24,453 connection–intervals where the preceding update tightened the ceiling, available export then increased, and the ceiling lay below both the current request and available export. The remaining connection–intervals gain a net 1.036 MWh, leaving a total loss of 4.435 MWh over the same 80 days. This partition locates the losses associated with output recovery after tightening; it does not isolate the effect of removing one participant's ceiling. Unit-gain ceilings return to capacity after any interval that does not qualify, so the reversal does not require persistent restrictions after the qualifying episode ends.

The complete-cohort case gains 10.957 MWh against no feedback and 10.958 MWh against the equal-total control. Both dependent intervals remain above zero. Customer voltages span 0.9090–1.0747 pu. This extends the comparison to 361 dates at the same load multiplier while jointly changing customer composition and calendar coverage.

S12.2 Ceiling memory across consecutive days

Retaining ceilings across midnight preserves the moderate-gain delivery advantage and the unit-gain loss. The comparison uses the same 80 assessment dates, which form ten calendar-contiguous blocks of 1–35 days. Within each block, the final ceiling update of one day supplies the initial ceiling of the next. Only block starts initialize ceilings at capacity. The three ECF gains remain 0.25, 0.5, and 1. Each trajectory generates its own equal-total control from its current admitted requests. The update without replay qualification also retains its own ceiling state at gain 0.5; no feedback requires no state transfer.

All raw requests, available output, mapping, network settings, and physical criteria remain unchanged.

No missing dates are synthesized or joined across calendar gaps. The static network model is initialized daily as before: the changed state is the request ceiling, not an internal storage or voltage-controller state. Thus the experiment isolates the role of ceiling initialization without changing the physical or behavioral model.

Supplementary Table 18: ECF delivery with ceilings retained across consecutive dates. Totals cover all 80 assessment days. The equal-total control is regenerated for each gain. All differences are calculated before rounding.

β_{ECF}	Delivered (MWh)	Minus no feedback (MWh)	Minus equal-total control (MWh)	Absolute change from daily-reset ECF (kWh)
0.25	239.308	1.345	1.345	0.0954
0.50	239.356	1.392	1.392	$< 10^{-5}$
1.00	234.920	−3.043	−0.537	$< 10^{-5}$

At gain 0.5, ECF delivers 239.356 MWh and the update without qualification delivers 232.933 MWh. Their difference is 6.423 MWh, positive on 75 days and negative on five. ECF exceeds no feedback on 44 days and delivers less on 32, with four unchanged. All three gains change aggregate ECF delivery by less than 0.1 kWh relative to daily initialization (Table 18). This small initialization effect does not remove variation between calendar blocks. Gain 0.5 loses delivery relative to no feedback in three blocks, whereas unit gain loses in every block (Table 19). In the longest 35-day block, the respective differences are +1.064 and −1.195 MWh.

Supplementary Table 19: Delivery differences in every consecutive-date block. Dates are in 2013 and all differences are in kWh. The last column compares gain-0.5 ECF with the same update without replay qualification, both retaining state across days.

Dates	Days	Gain 0.5 minus no feedback	Gain 1 minus no feedback	Gain 0.5 minus unqualified update
2–8 Jan.	7	−0.2	−336.8	438.9
12–25 Jan.	14	17.7	−593.1	922.3
2–4 Apr.	3	61.2	−156.9	413.3
9–10 Apr.	2	111.7	−36.0	4.3
14–15 Apr.	2	3.9	−17.4	191.5
22 Apr.	1	−122.4	−121.8	7.9
28 Apr.–1 May	4	−17.3	−198.7	354.9
6–12 May	7	207.9	−211.3	420.1
16 May–19 June	35	1063.6	−1194.7	3241.7
23–27 June	5	66.2	−176.4	427.8

The paired daily differences are retained alongside these block totals. Using the primary calendar-dependence model in (S81), the mean gain at 0.5 is 17.4 kWh per day against no feedback, with a 95% interval of 4.9–29.9 kWh. The covariance is positive for calendar-day separations of one to seven days and zero from eight days onward. The block totals separately show variation over longer consecutive periods.

Ceiling recovery explains why removing the daily reset need not produce persistent restriction. When the target remains at capacity, (S40) makes the remaining capacity deficit shrink by the factor $1-\beta_{\mathrm{ECF}}$ per update. This identity holds through midnight as well as within a day. Across all three gains, the largest midnight capacity deficits are 2.26 kW, 42.0 W, and zero, respectively. The longest uninterrupted runs in which a ceiling reduces an admitted request by more than 2 W are 16, 16, and four half-hours.

Runs continue across midnight and end at an unrestricted interval or a calendar gap. A short restriction is not necessarily harmless: unit gain has the shortest runs but the lowest aggregate delivery. Recovery duration alone therefore cannot rank the updates.

S12.3 Bounded records: qualification and later delivery

An error bound can make the estimated replay gain reliable without making the later update more productive. This comparison applies the lower record in (S34) during the feedback trajectory, rather than only evaluating it on fixed completed-state pairs. The prescribed radii are 1%, 5%, and 10% of the connection bound C_i. At each radius, all records have either the positive or negative bias. The point-record and lower-bound controls each evolve their own ceilings under the same raw requests, network, profiles, and gain 0.5. Each of the 80 days starts with $\psi_i = C_i$. An exact-record run adds a thirteenth setting and reproduces the primary ECF delivery of 239.356 MWh and all 483 qualifying intervals.

Suppressing the interval index, the raw record for participant i is $\max\{y_i, a_i + s\delta_i\}$, where $s \in \{-1, 1\}$ gives the common bias direction. Here $a_i = a_i^{\text{prof}}$ is the unperturbed modeled available export, used to generate test records and evaluate the outcome. It is not an input to the controller. The record is also checked against completed net metering. In the fixed-load, no-storage model, positive residual import implies zero available net export. Both controls therefore set the effective record to zero at importing connections. This shared consistency step uses net metering, not the unobserved available-export value. Original and effective records are retained. The radii are prescribed stress levels, not calibrated meter errors.

Both controls recompute the proportional replay award from the admitted requests and fixed network data. For the physical screen, both set connection-point active power equal to completed residual import minus candidate export, with household reactive demand fixed. Candidate export is the smaller of the replay award and effective record. Point-record control uses this delivery estimate in the recovery conditions. Lower-bound control uses the smaller of the replay award and $a_i^{\text{L}} = \max\{y_i, a_i^{\text{tel}} - \delta_i\}$ instead. The original 2-W thresholds and ceiling update are unchanged. Unperturbed availability is used after the decision for a separate recovery comparison and nonlinear AC check; neither result feeds back into the recorded update.

The bias endpoints have a useful analytic interpretation. For a connection without positive import, a positive bias gives $a_i^{\text{tel}} = a_i + \delta_i$ and hence $a_i^{\text{L}} = a_i$. At an importing connection both are zero after consistency checking. The positive-bias lower bound thus recovers the reference availability by construction. A negative bias instead gives $a_i^{\text{tel}} = \max\{y_i, a_i - \delta_i\}$ and $a_i^{\text{L}} = \max\{y_i, a_i - 2\delta_i\}$. The second subtraction accounts for the full admissible error interval, even when the realised error is already negative. It explains the greater conservatism of the lower-bound update under underestimation.

Every error setting retains a positive mean delivery difference relative to no feedback, with its date-adjusted 95% interval above zero (Table 20). This does not order the two record treatments. At 10% underestimation, the lower bound reduces total delivery by 269.1 kWh relative to the point estimate. Its mean daily difference is −3.36 kWh and the interval is entirely negative (Table 21). The other five intervals include zero. The intervals use paired dates, the covariance in (S81), and 10,000 antithetic wild replicates with fixed seed 176. They are pointwise intervals, without a multiple-comparison adjustment. The original primary comparison and its reported interval remain unchanged.

The qualification results explain a different aspect of this comparison. With positive bias, point-record control tightens after 11, 28, and 33 intervals whose reference recovery does not meet the required criteria. Lower-bound control has no such tightening in any error setting, as expected from the recovery bound. Under negative bias, it instead leaves more feasible recovery opportunities unused. At the largest radius it misses 258 of 496 opportunities along its trajectory, compared with 203 of 494 for point-record control (Table 22). These are different trajectories, not a common classification sample. A failed recovery condition may still have a positive change below 2 W; it does not necessarily indicate

Supplementary Table 20: Energy outcomes with bounded record errors over the same 80 dates. P denotes point-record control and L denotes delivery-lower-bound control. The bias magnitude equals the stated radius as a percentage of C_i. Energy totals and their differences are in MWh. The last column gives the date-adjusted 95% interval for the mean daily delivery difference in kWh. Both differences compare the stated control with no feedback.

Bias	Control	Delivered	Authorized	Unused	Delivery difference	Daily difference 95% interval (kWh)
Exact	L	239.356	329.696	90.341	1.392	[4.88, 29.93]
−1%	P	239.377	329.391	90.014	1.414	[6.01, 29.34]
−1%	L	239.423	329.485	90.062	1.459	[7.05, 29.43]
+1%	P	239.423	329.805	90.382	1.459	[6.30, 30.18]
+1%	L	239.356	329.696	90.341	1.392	[4.88, 29.93]
−5%	P	239.355	329.779	90.424	1.392	[6.17, 28.62]
−5%	L	239.178	329.728	90.550	1.215	[4.65, 25.72]
+5%	P	239.532	329.931	90.399	1.569	[7.61, 31.61]
+5%	L	239.356	329.696	90.341	1.392	[4.88, 29.93]
−10%	P	239.178	329.728	90.550	1.215	[4.65, 25.72]
−10%	L	238.909	329.357	90.448	0.946	[2.25, 21.39]
+10%	P	239.545	330.020	90.475	1.582	[7.79, 31.76]
+10%	L	239.356	329.696	90.341	1.392	[4.88, 29.93]

Supplementary Table 21: Delivery lower bound minus point-record control under the same bias. Mean differences and their date-adjusted 95% intervals are in kWh per day. Positive, negative, and unchanged counts refer to the 80 paired days.

Bias	Mean	95% interval	Positive	Negative	Unchanged
−1%	0.57	[−0.72, 1.85]	5	10	65
+1%	−0.84	[−2.24, 0.56]	4	6	70
−5%	−2.21	[−5.10, 0.67]	22	18	40
+5%	−2.20	[−4.43, 0.018]	5	14	61
−10%	−3.36	[−6.39, −0.34]	11	19	50
+10%	−2.37	[−4.77, 0.028]	6	14	60

a delivery loss. A feasible opportunity requires reference recovery and both replay-award and reference replay-delivery AC checks.

The lower record removes unsupported recovery decisions, but its additional conservatism under negative bias can also forgo useful updates. Reliable qualification and greater subsequent delivery are therefore distinct outcomes.

S13 Spatial and Capacity Variation

S13.1 Profile placement with unchanged uptake and chronology

The positive aggregate delivery contrast persists across five alternative placements of the same household profiles. Placement nevertheless changes the size and timing of the benefit. This comparison varies the spatial association of profiles with service connections while preserving their aggregate chronology and the export-participant locations.

Profile assignments. Mappings M01–M05 permute whole profiles within the export-participant and nonparticipant groups separately. A profile moves with its PV capacity and forecast-calibration history.

Supplementary Table 22: Recovery qualification along each control's own trajectory. Each row covers 3,840 half-hours. Unsupported means tightening despite failure of the reference delivery-recovery criteria. Missed counts use the feasible opportunities in the preceding column as their denominator.

Bias	Control	Tightenings	Unsupported	Feasible opportunities	Missed
Exact	L	483	0	483	0
−1%	P	459	0	486	27
−1%	L	436	0	485	49
+1%	P	491	11	480	0
+1%	L	483	0	483	0
−5%	P	362	0	485	123
−5%	L	291	0	494	203
+5%	P	507	28	479	0
+5%	L	483	0	483	0
−10%	P	291	0	494	203
−10%	L	238	0	496	258
+10%	P	512	33	479	0
+10%	L	483	0	483	0

The multiset of profiles in each group, including every repeated profile, remains unchanged. The same 418 of 1,255 connections retain export-participant status, and topology and phase connections are unchanged. Aggregate raw requests, available export, household load, and installed participant capacity are therefore the same under every assignment. Individual available export still varies by electrical location.

Within each group, receiving connections are ordered by load name. Donor profiles are ordered by a SHA-256 key formed from the mapping key, group and load name, with load name breaking ties. The five keys are 18001–18005. Ordered profiles are then assigned to ordered connections. The original placement remains the primary reference.

Matched trajectories. Each mapping uses the original 80 dates, gain 0.5, capacity initialization each day, exact profile-derived availability records, load multiplier 0.30, and unchanged PV scaling and network limits. The four controls are no feedback, previous-day persistence, ECF, and the equal-total control generated by that mapping's ECF trajectory. Table 23 gives all energy totals. For uncertainty estimates, the comparison unit is a paired network day within one mapping. The calendar-dependent wild bootstrap follows Section S10.3. Table 24 reports pointwise intervals for each mean difference.

Delivery and unused authorization. Relative to no feedback, the five 80-day delivery gains are 2.101, 0.865, 5.055, 1.361, and 2.948 MWh. Against the equal-total control they are 2.103, 0.865, 5.056, 1.360, and 2.949 MWh. Every corresponding pointwise mean-difference interval is above zero (Fig. 4). Previous-day persistence leaves much less authorization unused but also delivers less than ECF under every mapping (Table 23).

M02 provides a further distinction between unused permission and lost delivery. Relative to no feedback, its authorization rises by 1.217 MWh and unused authorization rises by 0.353 MWh, leaving a net delivery gain of 0.865 MWh. The identity $\Delta E_Y = \Delta E_X - \Delta(E_X - E_Y)$ explains this result without requiring unused authorization to decrease. M02 has 24 positive, 43 negative, and 13 unchanged days against no feedback. Its positive total comes from larger gains on fewer days, rather than a daily advantage. The remaining four mappings reduce aggregate unused authorization relative to both no feedback and the equal-total control.

Update selection. The numbers of qualifying half-hours are 662, 406, 856, 463, and 798 for M01–M05, respectively, each out of 3,840 assessed half-hours. Thus relocating the same profiles changes how often usable recovery is found, while retaining a positive aggregate delivery contrast under all five assignments.

Supplementary Table 23: Energy totals for five profile placements over the same 80 dates. All energy columns are in MWh. Unused authorization equals authorized minus delivered energy before rounding. Negative source exchange denotes net import. Persistence is the previous-day persistence control.

Mapping	Control	Authorized	Delivered	Unused	Source exchange
M01	No feedback	308.613	226.378	82.235	-272.260
M01	ECF	309.432	228.480	80.953	-270.181
M01	Equal-total	308.583	226.376	82.207	-272.262
M01	Persistence	216.472	193.063	23.409	-305.430
M02	No feedback	163.945	118.588	45.357	-379.394
M02	ECF	165.162	119.452	45.710	-378.534
M02	Equal-total	163.936	118.587	45.349	-379.395
M02	Persistence	123.241	111.558	11.683	-386.468
M03	No feedback	334.542	245.413	89.129	-253.390
M03	ECF	338.164	250.469	87.695	-248.393
M03	Equal-total	334.505	245.412	89.093	-253.391
M03	Persistence	240.465	211.730	28.735	-286.947
M04	No feedback	381.071	270.372	110.700	-228.766
M04	ECF	382.011	271.732	110.278	-227.424
M04	Equal-total	381.055	270.372	110.683	-228.766
M04	Persistence	259.848	229.406	30.442	-269.482
M05	No feedback	256.245	187.677	68.568	-310.653
M05	ECF	258.727	190.625	68.101	-307.730
M05	Equal-total	256.235	187.676	68.559	-310.654
M05	Persistence	178.468	160.345	18.122	-337.900

S13.2 PV capacity at fixed participant locations

ECF retains a positive aggregate delivery advantage across five PV capacity levels, but the advantage does not grow in proportion to installed capacity. This comparison changes generation capacity at the same participant locations, complementing the profile-placement comparison in Section S13.1. It also shows that higher delivery can accompany either more or less unused authorization.

Capacity and matched operating conditions. The capacity levels are 0.50, 0.75, 1.00, 1.25, and 1.50 times the primary scale, corresponding to 2.045–6.135 MW across the 418 export participants. Gross PV profiles and connection bounds C_i use the same factor. Available export is then recalculated after subtracting the unchanged load; it is not obtained by multiplying the original net surplus. The forecast rule retains its 90th-percentile increment, recalculated from the scaled 2010–2011 profiles. Previous-day persistence uses the corresponding scaled profile without that increment.

All levels retain the same participant locations, profile assignment, phase connections, 80 dates, load multiplier 0.30, feedback gain 0.5, and daily capacity initialization. The allocation search and physical checks are unchanged, including the 1.07-pu authorization ceiling, 0.90–1.10-pu delivery range, and native transformer limits. No feedback, previous-day persistence, ECF, and each ECF trajectory's

Supplementary Table 24: ECF minus each control within each profile placement. Daily means and their pointwise dependence-adjusted 95% intervals are in kWh per network day. The last column counts positive, negative, and unchanged days, in that order, and sums to 80 in every row. The intervals describe mean differences, not the spread of daily outcomes.

Mapping	Comparator	Total (MWh)	Daily mean	Lower	Upper	Days (+/ − /0)
M01	No feedback	+2.101	26.27	14.32	38.22	49/31/0
M01	Equal-total	+2.103	26.29	14.35	38.23	49/31/0
M01	Persistence	+35.416	442.70	397.45	487.95	80/0/0
M02	No feedback	+0.865	10.81	3.50	18.12	24/43/13
M02	Equal-total	+0.865	10.81	3.50	18.12	29/42/9
M02	Persistence	+7.894	98.68	66.49	130.88	61/19/0
M03	No feedback	+5.055	63.19	45.47	80.92	67/13/0
M03	Equal-total	+5.056	63.21	45.47	80.94	67/13/0
M03	Persistence	+38.739	484.24	432.01	536.46	80/0/0
M04	No feedback	+1.361	17.01	11.88	22.13	60/17/3
M04	Equal-total	+1.360	17.01	11.88	22.13	60/17/3
M04	Persistence	+42.326	529.08	444.49	613.66	80/0/0
M05	No feedback	+2.948	36.85	26.48	47.22	76/4/0
M05	Equal-total	+2.949	36.87	26.48	47.25	76/4/0
M05	Persistence	+30.280	378.50	335.23	421.77	80/0/0

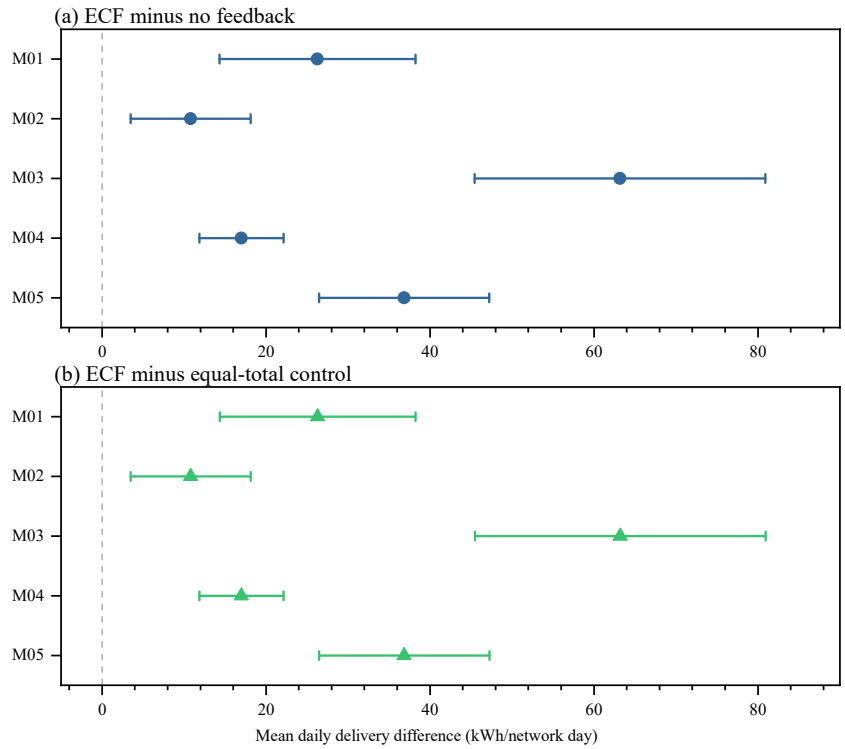
own equal-total control are compared within each capacity level. The original factor 1.00 remains the primary reference.

Delivery, scarcity, and authorization. Across all five levels, ECF gains 1.392–1.879 MWh over no feedback and 1.392–1.884 MWh over the equal-total control. The corresponding mean-difference intervals remain above zero. The primary results are given in Table 12; Table 26 reports the four added levels. Every level nevertheless includes days on which ECF delivers less than either control.

The fraction of no-feedback half-hours with a network-limited request scale rises from 37.0% to 46.0% as capacity increases. ECF qualification rises from 11.5% to 13.3%, but its delivery advantage is not monotonic (Table 25). The gain falls from 0.813 to 0.273 MWh per installed MW over the 80 dates. Thus more frequent constrained allocation does not produce a proportionally larger feedback benefit. The constrained-request fraction counts half-hours with a reduced proportional allocation factor under the stated search.

At factor 0.50, authorization increases by 2.262 MWh and unused authorization by 0.599 MWh relative to no feedback. Their difference gives the 1.662-MWh delivery gain. Factors 0.75 and 1.50 also increase unused authorization, whereas factors 1.00 and 1.25 reduce it. The energy identity accommodates a positive delivery difference whether unused authorization increases or decreases. All component differences compare the same pair of control trajectories over the same dates. Previous-day persistence leaves less authorization unused at every level but delivers 47.52–51.94 MWh less than ECF (Table 27).

Uncertainty. Each added level uses 80 paired network days. The pointwise 95% intervals use the calendar-dependent wild bootstrap in Section S10.3 (random key 184).



Supplementary Fig. 4: Profile placement changes the mean delivery advantage. Markers show ECF-minus-control mean modeled delivery per network day, and horizontal bars show pointwise 95% intervals adjusted for calendar dependence. (a) ECF minus no feedback. (b) ECF minus the equal-total control. Each row uses the same 80 dates under one fixed profile assignment. The zero line denotes no mean difference. Identical axis scales allow the two comparisons to be read directly; daily losses remain included in each mean.

Supplementary Table 25: PV capacity, constrained requests, and the ECF-minus-no-feedback energy balance over 80 dates. Capacity is installed participant PV in MW. Limited and Qualified are percentages of all 3,840 half-hours: a reduced allocation factor without feedback and a qualifying ECF replay, respectively. Authorization, unused authorization, and delivery changes are in MWh. Gain/MW is additional delivered MWh per installed participant MW over these dates. Differences use unrounded totals.

Factor	Capacity	Limited	Qualified	Authorization	Unused	Delivery	Gain/MW
0.50	2.045	37.0	11.5	2.262	0.599	1.662	0.813
0.75	3.067	41.9	11.9	1.882	0.003	1.879	0.612
1.00	4.090	44.4	12.6	1.206	−0.186	1.392	0.340
1.25	5.112	45.3	12.8	1.342	−0.147	1.488	0.291
1.50	6.135	46.0	13.3	1.867	0.189	1.678	0.273

S13.3 Variation across low-voltage sections of the parent feeder

ECF delivers more than the equal-total control in all 13 retained LV sections. Against no feedback, 12 differences are positive and one is negative. Spatial targeting therefore improves on uniform reduction throughout this panel, but it does not always improve on leaving requests unchanged.

The public collection provides executable models for its nested LV sections. We retain sections with at least 20 mapped load connections and five export participants: LV9, LV10, LV11, LV12, LV15,

Supplementary Table 26: Paired delivery comparisons at the four additional PV capacities. Differences are ECF minus the named control. Mean and pointwise 95% interval are in kWh per network day; Total is in MWh over 80 dates. The last column counts positive, negative, and unchanged daily differences, respectively. Equal-total denotes the matched proportional request reduction, and Persistence denotes the previous-day request without the forecast increment.

Factor	Control	Mean	95% interval	Total	+/ − /0 days
0.50	No feedback	20.78	[11.19, 30.37]	1.662	51/28/1
	Equal-total	21.31	[11.82, 30.79]	1.705	51/28/1
	Persistence	594.06	[552.14, 635.97]	47.524	80/0/0
0.75	No feedback	23.48	[11.06, 35.91]	1.879	49/31/0
	Equal-total	23.55	[11.13, 35.98]	1.884	49/31/0
	Persistence	622.18	[585.45, 658.91]	49.774	80/0/0
1.25	No feedback	18.60	[5.53, 31.67]	1.488	47/29/4
	Equal-total	18.61	[5.54, 31.68]	1.489	48/30/2
	Persistence	639.03	[604.07, 673.98]	51.122	80/0/0
1.50	No feedback	20.97	[6.83, 35.11]	1.678	46/32/2
	Equal-total	20.97	[6.83, 35.11]	1.678	47/32/1
	Persistence	649.23	[614.92, 683.55]	51.939	80/0/0

Supplementary Table 27: Energy totals at all five PV capacities over the same 80 dates. All energy columns are in MWh. Unused is authorized minus delivered energy; negative source exchange denotes net upstream import. The factor-1.00 rows repeat the unchanged primary reference. All four controls pass the specified physical checks at every level.

Factor	Control	Authorized	Unused	Delivered	Source exchange
0.50	No feedback	298.696	98.512	200.183	−303.934
	Persistence	184.406	30.085	154.321	−349.520
	ECF	300.957	99.111	201.846	−302.289
	Equal-total	298.406	98.264	200.141	−303.975
0.75	No feedback	318.310	94.121	224.190	−276.706
	Persistence	202.295	26.001	176.294	−324.304
	ECF	320.192	94.124	226.068	−274.848
	Equal-total	318.133	93.949	224.184	−276.712
1.00	No feedback	328.490	90.526	237.963	−260.959
	Persistence	212.296	23.267	189.030	−309.584
	ECF	329.696	90.341	239.356	−259.582
	Equal-total	328.483	90.519	237.964	−260.959
1.25	No feedback	335.183	87.949	247.234	−250.321
	Persistence	218.954	21.354	197.600	−299.639
	ECF	336.524	87.802	248.722	−248.849
	Equal-total	335.164	87.931	247.233	−250.321
1.50	No feedback	340.650	86.634	254.017	−242.514
	Persistence	223.567	19.811	203.756	−292.454
	ECF	342.517	86.823	255.694	−240.854
	Equal-total	340.649	86.632	254.017	−242.514

LV17, LV22, LV24, LV25, LV26, LV29, LV30, and LV32. Each native OpenDSS master model is solved separately. All sections use the common chronology and scenario of the parent collection. Their energy differences describe separate section models; the full-feeder difference is evaluated in the coupled MV/LV model.

Every selected section uses the primary 80-day assessment calendar, profile-to-service mapping, load scale, allocation-stage upper-voltage ceiling, request model, and $\beta_{\mathrm{ECF}} = 0.5$. Table 28 reports every outcome. All 13 sections contain qualifying intervals. LV26 alone has a negative total relative to no feedback (−3.7 kWh), but gains 71.2 kWh relative to the equal-total control. Modeled-delivery voltages range from 0.9136 to 1.0952 pu across the panel. No modeled-delivery state exceeds its native transformer nameplate.

Supplementary Table 28: Within-collection low-voltage-section outcomes over the 80 assessment days. Differences are ECF minus the stated control in kWh. The final column indicates whether all specified allocation and modeled-delivery checks pass across the four controls.

Section	Connections	Participants	ECF − no feedback	ECF − equal-total control	Qualifying intervals	All checks pass
LV9	123	35	844.7	873.8	580	Yes
LV10	94	33	371.5	454.5	374	Yes
LV11	110	41	511.7	545.2	360	Yes
LV12	121	36	1304.5	1311.6	595	Yes
LV15	46	12	198.2	200.3	237	Yes
LV17	141	48	107.0	287.1	308	Yes
LV22	21	9	205.0	206.1	348	Yes
LV24	121	44	947.0	985.5	329	Yes
LV25	112	39	718.0	783.7	505	Yes
LV26	94	31	−3.7	71.2	101	Yes
LV29	41	17	54.3	83.7	73	Yes
LV30	151	51	1213.5	1261.4	371	Yes
LV32	22	7	95.5	99.3	273	Yes

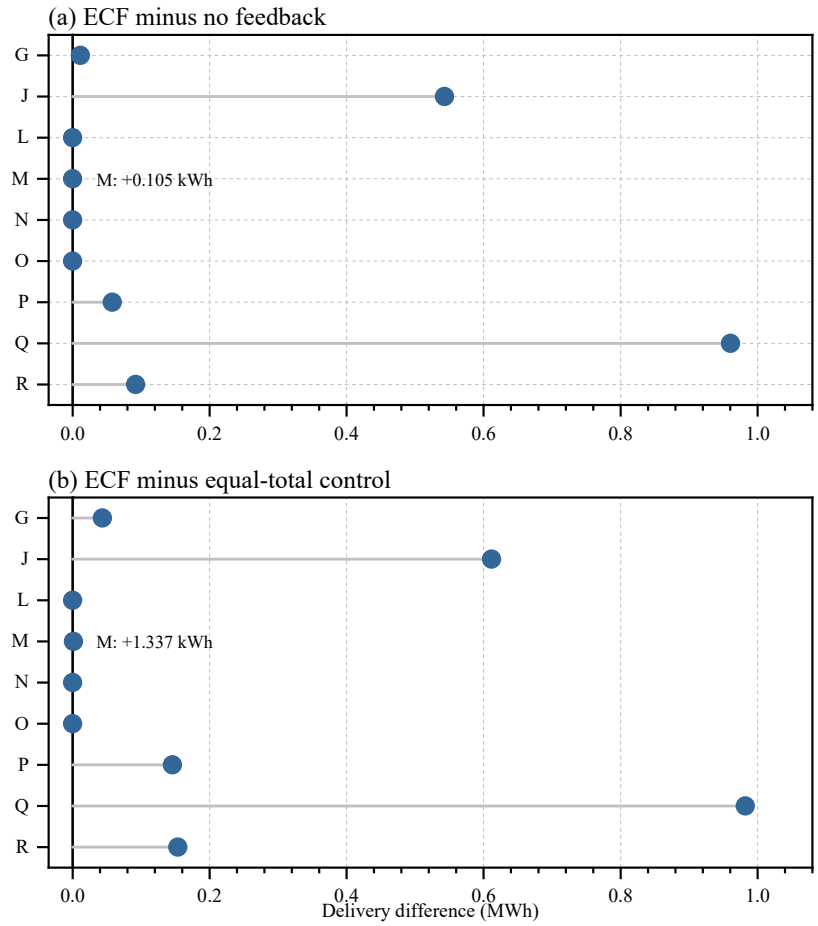
S13.4 Variation across representative public low-voltage models

The representative-model panel changes the published LV topology rather than the location within one parent feeder. Six of nine eligible models contain qualifying intervals and show positive aggregate delivery differences against both controls. Models L, N, and O have no qualifying intervals and no delivery change.

The panel begins with the 23 representative Australian LV models released by the National Low-Voltage Feeder Taxonomy Study[13]. Eligibility is determined from structure and a connection-voltage check with mapped household injections set to zero. A retained model must converge and satisfy the voltage criterion on every declared connection phase, and it must have a usable native transformer nameplate. It must also contain at least 20 mapped load premises and five export participants.

The phase-preserving representation retains each published load element as one allocation premise. It preserves the one-, two-, or three-phase connection and checks voltage on every declared phase. These criteria select G, J, L, M, N, O, P, Q, and R. Models A–F, H, I, K, and S lack a native transformer with a usable nameplate. Models T, U, and V do not meet the size rule. Model W violates the zero-injection connection-voltage criterion.

Each selected model receives the same 80 observed dates, deterministic profile assignment, request construction, load scale, allocation-stage upper-voltage ceiling, four controls, and $\beta_{\mathrm{ECF}} = 0.5$. The published model supplies each topology, its connection structure, and its native equipment data. Table 29 reports every selected model separately.



Supplementary Fig. 5: Delivery differences across nine published representative low-voltage models. Each marker is the 80-day ECF-minus-control total for one model; horizontal stems connect zero to that total. (a) ECF minus no feedback. (b) ECF minus the equal-total control. Six models have positive totals; L, N, and O have no qualifying interval and zero difference.

Supplementary Table 29: Results for the public representative-LV-model panel over the 80 assessment days. Differences are ECF minus the named control in kWh. Every model passes the specified allocation, customer-voltage, and native transformer checks. Small positive differences are retained rather than rounded to zero.

Model	Conn.	Premise phase counts	ECF − no feedback	ECF − equal-total control	Qualifying intervals	All physical checks pass	Min. voltage (pu)
G	59	1	11.4	43.5	69	Yes	0.9749
J	87	1	543.0	611.7	328	Yes	0.9412
L	36	1	0.0	0.0	0	Yes	0.9792
M	58	1,3	0.1	1.3	2	Yes	0.9662
N	63	1,3	0.0	0.0	0	Yes	0.9734
O	83	1,2	0.0	0.0	0	Yes	0.9823
P	109	1,2,3	58.1	145.9	85	Yes	0.9734
Q	109	1,3	960.5	982.1	685	Yes	0.9384
R	144	1,2	92.1	153.7	97	Yes	0.9594

S14 Participant Outcomes and Scope

S14.1 Individual delivery changes under shared constraints

The primary aggregate gain is shared unevenly. Relative to no feedback, 330 of the 418 participant connections gain cumulative delivery and 88 lose delivery over the 80 assessed days. The positive changes sum to 1.557 MWh and the negative changes to −0.165 MWh, leaving the reported net gain of 1.392 MWh. The median individual change is 2.6 kWh and the largest loss is 7.3 kWh. The equal-total comparison has the same participant counts and a similar distribution (Table 30).

Each individual difference sums ECF-minus-control delivery over the same half-hours. The comparison with the same update without replay qualification instead gives 417 positive individual totals and one negative total of −1.17 kWh. Thus retaining qualification benefits more participants than updating after every unused award, but does not make every participant better off than no feedback.

Positive cumulative delivery also does not imply a benefit on every day. All 418 participants have at least one negative day against no feedback or the equal-total control. The longest uninterrupted run of negative daily differences is five days against either control. A missing calendar date breaks a run; separated assessment days are not treated as consecutive. The corresponding maximum against the update without qualification is seven days. A positive cumulative result can therefore coexist with several consecutive days of lower delivery at the same connection.

Supplementary Table 30: Individual delivery differences in the primary 80-day comparison. All differences are ECF minus the named control. Energy rows are in kWh per participant unless stated otherwise. Counts cover all 418 participant connections. The last column uses the same ceiling update without replay qualification. Day-run statistics break at gaps in the assessment calendar.

	No feedback	Equal-total	No qualification
Positive cumulative difference	330	330	417
Negative cumulative difference	88	88	1
Unchanged cumulative difference	0	0	0
Minimum individual difference	−7.31	−7.31	−1.17
5th percentile	−2.83	−2.83	3.07
25th percentile	0.43	0.43	7.70
Median	2.60	2.60	12.57
75th percentile	5.09	5.09	19.86
95th percentile	13.52	13.52	36.06
Maximum individual difference	30.50	30.50	124.38
Participants with any negative day	418	418	417
Maximum negative-day count	34	36	36
Longest consecutive negative-day run	5	5	7

S14.2 Requests adjusted from participants' own delivery

We test whether delivery comparisons change when participants revise their requests after observing their own completed delivery. Each of the 418 export participants chooses among five multipliers, $m_k \in \{0.50, 0.75, 1.00, 1.25, 1.50\}$, where $k = 1, \ldots, 5$. These choices allow both smaller and larger requests. They are distinct from the optimal larger-request multiplier $\gamma_i^\star$ in the controlled study. Let $r_i^{\text{base}}(t)$ denote the capacity-capped, history-based request used by the primary public-data simulation before any feedback ceiling is applied. If participant i selects candidate k, its capacity-capped request is

$$r_i^{\text{cap}}(t) = \min\{C_i, m_k r_i^{\text{base}}(t)\}. \tag{S94}$$

The applicable request ceiling and the original allocation rule then determine admitted requests and awards. The network, profile mapping, PV capacity, feedback formula, and nominal gain remain unchanged.

We use an Exp3 request-selection update[14]. After an interval, a participant increases the selection weight of its chosen candidate in proportion to its normalized delivery reward. Let $w_{ik}^{\mathrm{req}}(t) > 0$ denote participant i's selection weight for candidate k, and let $p_{ik}^{\mathrm{req}}(t)$ denote its selection probability. The weights start at one, and selection uses

$$p_{ik}^{\mathrm{req}}(t) = 0.90 \frac{w_{ik}^{\mathrm{req}}(t)}{\sum_{h=1}^{5} w_{ih}^{\mathrm{req}}(t)} + \frac{0.10}{5}. \tag{S95}$$

After delivery is completed, only the selected candidate's weight is updated:

$$w_{ik}^{\mathrm{req}}(t+1) = w_{ik}^{\mathrm{req}}(t) \exp\left[\frac{0.10}{5 p_{ik}^{\mathrm{req}}(t)} \frac{y_i(t)}{C_i} \right]. \tag{S96}$$

Other weights are unchanged. Computation uses logarithmic weights and a common row normalization, which leaves the probabilities unchanged. Zero baseline requests produce zero delivery and no weight update. The reward is the participant's own delivered power divided by its connection bound, consistent with the primary zero-shortfall-cost setting. Request selection does not use current available export, other participants' private information, or replay outcomes. A comparison with no weight update retains the same initial probabilities and random draws but leaves all weights unchanged. It still samples request multipliers; "fixed" refers to the weights, not the requests.

No feedback, ECF, and the same update without qualification each have their own delivery histories and evolving selection weights. Their contrasts therefore include subsequent request responses. The equal-total shadow instead follows ECF's history and current capacity-capped requests, removing the same total request at that interval. Its contrast retains the conditional comparison of where requests are reduced; it is not the trajectory of an independently adapting uniform controller.

Both selection weights and ceilings persist across midnight within each of the ten consecutive-date blocks. They are initialized again after calendar gaps; the longest block spans 35 days. Each behavior mode uses seeds 2001, 2002, and 2003, fixed before evaluation. We retain every seed and date, first averaging the three seeds within each date for the main paired comparisons. Uncertainty is computed over the original 80 dates with the same calendar-dependent procedure used for the other public-data comparisons. Seed-specific and block-level contrasts accompany the date-averaged results.

This participant model tests experience-based request adjustment. It does not model strategies that anticipate future ceilings or manipulate availability records. Exp3's regret guarantees are not transferred to the coupled allocation and feedback system.

Updating request-selection probabilities preserves ECF's aggregate delivery advantage (Fig. 6 and Table 31). Relative to no feedback, the mean gain is 18.1 kWh per day (95% interval 10.5–25.6), or 1.44 MWh over the 80 dates. The gain over the same update without qualification is 56.6 kWh per day (46.0–67.3), or 4.53 MWh. All three seeds give positive mean contrasts, although individual dates and consecutive-date blocks can lose delivery. ECF exceeds the conditional equal-total shadow by 1.50 MWh. This last comparison holds ECF's request history fixed; the first two include the separate histories produced by each controller.

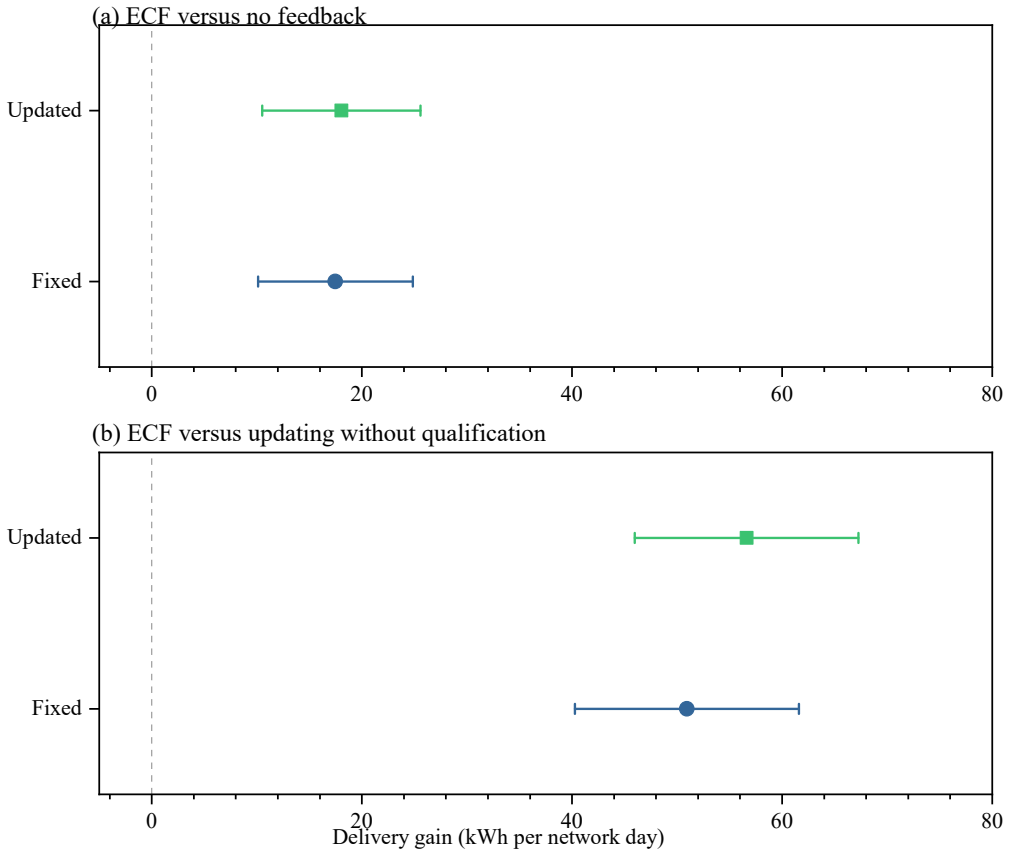
The energy balance explains why lower unused authorization is not the sole source of benefit (Table 32). Against no feedback, ECF grants 1.31 MWh more authorization and leaves 0.14 MWh less unused, together delivering 1.44 MWh more. Against updating without qualification, it grants 20.99 MWh more authorization and leaves 16.46 MWh more unused. The remaining 4.53 MWh reaches delivery. Thus a controller can suppress unused authorization more aggressively while also removing useful awards.

Updating these probabilities changes the requests passed to allocation. Their time integral under ECF is 1656.33 MWh, compared with 1629.26 MWh at fixed probabilities, after averaging the three

seeds. This integral measures requests over time, not awarded or delivered energy. To isolate the effect of weight updates, the comparison keeps the candidate requests and random draws unchanged; only the weight update differs.

Table 33 compares ECF's advantage between these two behavior modes. Updating request-selection probabilities increases its daily advantage over no feedback by 0.58 kWh, with a 95% interval of −0.58 to 1.73. For the conditional equal-total shadow, the increase is 1.22 kWh (−0.02 to 2.45). Neither interval resolves an increase above zero. In contrast, the advantage over updating without qualification increases by 5.68 kWh per day (2.54–8.83). The results therefore support persistence of ECF's mean advantage under experience-based request adjustment, rather than a general claim that adaptation amplifies it.

Table 34 retains the positive and negative block contrasts. All date- and seed-level results accompany the plotting data. The pointwise 95% intervals use the covariance in (S81), with 10,000 antithetic draws and seed 189; they do not adjust for multiple comparisons.



Supplementary Fig. 6: Delivery gains with fixed and updated request-choice probabilities. ECF is compared with (a) no feedback and (b) the same ceiling update without replay qualification. Blue circles denote fixed probabilities; green squares denote probabilities updated from each participant's own completed delivery. Both modes randomly select among the same candidate requests. Markers show mean daily differences after averaging the three seeds within each of the 80 dates. Horizontal bars show pointwise 95% intervals accounting for calendar dependence. The panels share the same horizontal scale. Each controller maintains its own request and delivery history. Table 33 reports the paired change in ECF's advantage between the two request-choice modes.

Supplementary Table 31: Delivery gains under updated and fixed request-choice probabilities. All differences are ECF minus the named control. Mean and pointwise 95% interval are in kWh per network day; Total is in MWh over 80 dates. Mean-seed rows average the three participant-response seeds within each date before aggregation. The last column counts positive, negative, and unchanged dates. The equal-total shadow follows ECF's request history.

Probabilities	Control	Seed	Mean	95% interval	Total	+/ − /0 days
Updated	No feedback	2001	13.21	[5.61, 20.81]	1.057	43/37/0
Updated	No feedback	2002	21.66	[11.70, 31.61]	1.733	51/28/1
Updated	No feedback	2003	19.29	[9.98, 28.60]	1.543	51/29/0
Updated	No feedback	Mean	18.05	[10.52, 25.59]	1.444	54/26/0
Updated	Without qualification	2001	57.49	[46.59, 68.38]	4.599	71/9/0
Updated	Without qualification	2002	56.27	[43.10, 69.44]	4.502	67/13/0
Updated	Without qualification	2003	56.11	[43.56, 68.66]	4.489	71/9/0
Updated	Without qualification	Mean	56.62	[45.98, 67.26]	4.530	73/7/0
Updated	Equal-total shadow	2001	13.83	[6.05, 21.60]	1.106	44/36/0
Updated	Equal-total shadow	2002	22.58	[12.52, 32.64]	1.806	51/28/1
Updated	Equal-total shadow	2003	19.67	[10.36, 28.98]	1.574	53/26/1
Updated	Equal-total shadow	Mean	18.69	[11.02, 26.37]	1.495	55/25/0
Fixed	No feedback	2001	13.62	[5.10, 22.13]	1.089	48/32/0
Fixed	No feedback	2002	21.04	[11.17, 30.90]	1.683	49/30/1
Fixed	No feedback	2003	17.78	[9.58, 25.98]	1.422	50/29/1
Fixed	No feedback	Mean	17.48	[10.12, 24.84]	1.398	54/26/0
Fixed	Without qualification	2001	53.26	[42.24, 64.28]	4.261	69/11/0
Fixed	Without qualification	2002	50.35	[36.27, 64.44]	4.028	64/16/0
Fixed	Without qualification	2003	49.20	[35.95, 62.44]	3.936	68/12/0
Fixed	Without qualification	Mean	50.94	[40.27, 61.60]	4.075	74/6/0
Fixed	Equal-total shadow	2001	13.62	[5.10, 22.13]	1.089	48/32/0
Fixed	Equal-total shadow	2002	21.04	[11.18, 30.90]	1.683	49/30/1
Fixed	Equal-total shadow	2003	17.77	[9.57, 25.97]	1.422	50/29/1
Fixed	Equal-total shadow	Mean	17.48	[10.12, 24.84]	1.398	54/26/0

Supplementary Table 32: Energy totals under the two request-choice modes, averaged over the three seeds. All entries are MWh over the same 80 dates. Unused equals authorized minus delivered energy; negative source exchange denotes net upstream import. Differences are calculated before rounding.

Probabilities	Control	Authorized	Unused	Delivered	Source exchange
Updated	No feedback	311.55	85.90	225.65	-273.21
Updated	ECF	312.86	85.76	227.09	-271.78
Updated	Without qualification	291.86	69.30	222.56	-276.27
Updated	Equal-total shadow	311.44	85.85	225.60	-273.26
Fixed	No feedback	308.81	84.78	224.03	-274.82
Fixed	ECF	310.08	84.66	225.43	-273.43
Fixed	Without qualification	290.06	68.71	221.35	-277.48
Fixed	Equal-total shadow	308.79	84.77	224.03	-274.82

Supplementary Table 33: Change in ECF's delivery advantage when request-choice probabilities are updated rather than fixed. Each entry is the updated-probability ECF-minus-control contrast minus its fixed-probability counterpart, with seeds averaged within each date. Mean and pointwise 95% interval are in kWh per day; Total is MWh over 80 dates.

Control	Mean	95% interval	Total	$+/-/0$ days
No feedback	0.58	[-0.58, 1.73]	0.046	36/44/0
Without qualification	5.68	[2.54, 8.83]	0.455	57/23/0
Equal-total shadow	1.22	[-0.02, 2.45]	0.097	38/42/0

Supplementary Table 34: Delivery contrasts within every consecutive-date block. Entries are kWh, ECF minus each control, averaged over three seeds. Block numbering follows chronological order in Section S12.2. Request-selection weights and ceilings persist within a block and restart after calendar gaps; gains need not be positive in every block.

Probabilities	Block	Days	No feedback	Without qualification	Equal-total shadow
Updated	1	7	35.02	122.20	38.27
Updated	2	14	165.85	634.78	163.67
Updated	3	3	64.87	367.70	65.54
Updated	4	2	93.28	-13.94	92.85
Updated	5	2	-18.23	35.51	-16.58
Updated	6	1	0.02	89.51	-1.31
Updated	7	4	1.72	258.17	3.96
Updated	8	7	215.44	322.15	215.98
Updated	9	35	802.90	2297.81	849.63
Updated	10	5	83.43	415.83	83.47
Fixed	1	7	45.17	116.09	45.15
Fixed	2	14	133.45	543.69	133.58
Fixed	3	3	65.71	362.56	65.70
Fixed	4	2	92.64	-16.96	92.63
Fixed	5	2	-15.66	35.30	-15.66
Fixed	6	1	0.84	90.30	0.83
Fixed	7	4	2.38	251.95	2.35
Fixed	8	7	230.79	323.76	230.80
Fixed	9	35	763.97	1959.77	763.96
Fixed	10	5	78.84	408.59	78.83

S14.3 Shared scope and physical-check coverage

Sections S11–S14 address distinct explanations for the delivery gain: request placement, update selection, the strength and persistence of the ceiling response, and variation in records and participants. The comparisons use the statistical units defined at the start of Section S11 and the customer-voltage and native-transformer criteria in Section S10. All reported public-study trajectories satisfy these criteria. Table 35 collects their coverage so that record counts are distinguished from statistical sample sizes.

Supplementary Table 35: Coverage of the public-study physical checks. An interval–control record contains the allocation and modeled-delivery states for one control in one half-hour. Replay entries count additional completed-state comparisons or states, as specified. All listed checks pass. Rows overlap where they reuse primary trajectories and must not be summed as independent observations. A dash indicates that no separate replay count is tabulated here, not that replay was omitted. Intermediate allocation-search candidates are excluded.

Comparison	Interval–control records	Additional replay coverage
Primary four controls (S11.1)	15,360	–
Same update with/without qualification (S11.2)	7,680	–
Twelve random schedules (S11.3)	46,080	Replay-free controls
Complete-customer cohort (S12.1)	69,312	–
Cross-day ceiling memory (S12.2)	26,880	11,520 ECF replay states
Bounded-record controls (S12.3)	49,920	49,920 record-based states and 49,920 reference states
Five profile placements (S13.1)	76,800	19,200 replay comparisons
Four additional PV scales (S13.2)	61,440	15,360 replay comparisons
Thirteen nested LV sections (S13.3)	199,680	–
Participant request adjustment (S14.2)	92,160	23,040 replay states

The primary, gain/record, complete-cohort, nested-section and representative-model suite contains 30 cases and 514,752 records (15,360 primary and 499,392 other records). This overlapping suite count is retained separately from the additional comparisons above. Table 29 reports the nine representative models individually.

Cross-day recursion checks cover 15,360 ceiling updates and 280 day-boundary transfers; the first day of each block agrees with its daily-reset trajectory. The capacity-target recovery identity was checked over 10,117 updates. Bounded-record tests verify the recursion, record bounds and energy identities at every connection and interval. The mapping and PV-scale replay endpoints give the same physical status on separate nonlinear AC evaluation, and the participant-adjustment states pass independent AC re-evaluation. Power-flow call counts in Table 15 include replay when required; they are operation counts, not execution time or record-validation costs.

The public-study physical checks cover customer voltages and transformer loadings. Line ampacities are unavailable, and the bounded-record study checks specified states rather than every state within an error interval. Representative models provide distinct published structures; nested sections share one parent feeder. Mapping and PV-size comparisons retain the participant adoption pattern. Continuous-state results use observed consecutive-date blocks; the covariance model in (S81) does not bound the duration of feedback dependence. The main Discussion brings these limits together with the operational requirements for linked records and modeled controller responses.

Taken together, the comparisons show why delivered electricity and unused authorization must be assessed separately. ECF can preserve useful awards while reducing requests at selected connections and times. Its benefit depends on both the completed-interval comparison and the subsequent ceiling response: a ceiling that follows low completed delivery too closely can restrict later output. This connects the derivations and simulations to the main paper's objective of using existing export access more effectively.

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